

FOCUS ON

Combined Sciences

New Curriculum

LEARNER'S BOOK

Elloise du Toit • Fionna Willmot • Robin Carter



college press

FORM

3

FOCUS ON

Combined Science

Learner's Book
Form 3



9

7

Contents

Biology

Topic 1 Structure and function of a cell	1
Specialised cells	1
Topic 2 Nutrition in plants	8
Photosynthesis	8
Topic 3 Nutrition in humans	24
The human alimentary canal	24
The functions of the parts of the alimentary canal	26
Food tests	34
Topic 4 Respiratory system	41
Inhaled and exhaled air	41
Topic 5 Transport in plants	47
Transport in plants	47
Topic 6 Transport system in humans	56
Blood circulation	56
Topic 7 Reproductive systems in plants	59
Wind and insect pollinated flowers	59
Germination	63
Topic 8 Reproductive systems in humans	69
The male reproductive system	69
The female reproductive system	70
Topic 9 Health and diseases	79
Sexually transmitted infections (STIs)	79
Dangerous infectious diseases in Africa	82
Substance abuse	84

Physics

Topic 1 Data presentation	94
Presenting data	94
Pie charts	94
Line graphs	97
Topic 2 Measurement	100
Measurement of physical quantities	100
Determining the density of liquids	100

Topic 3 Force	114
Weight and mass	114
Momentum	115
Inertia	116
Newton's laws of motion	117
Machines	117
Topic 4 Energy	124
Petrol and diesel engines	135
Topic 5 Energy	143
Heat transfer	143
Topic 6 Magnetism	156
Electromagnetism	156
Motor effect	161
Generator effect	166
Topic 7 Electricity	176
Static electricity	177
Lightning conductor	183
Current electricity	186
Ohm's law	187
Electrical power and energy	198

Chemistry

Topic 1 Separation	207
Distillation	207
Topic 2 Matter	214
The atom	215
The periodic table of element	217
Nuclide notation	219
Isotopes	221
Electronic configurations	222
Bonding	223
Covalent bonding	225
Relative molecular mass and relative formula mass	231
The mole concept and the Avogadro constant ..	232
Empirical formula	236
Concentration	239

Topic 3 Acids and bases.....244

The pH scale and universal indicator..... 244

Acid-base reactions..... 247

Representing chemical reactions..... 248

Topic 4 Industrial processes256

Fractional distillation..... 257

Electrolysis..... 259

Electroplating..... 265

Topic 5 Oxidation and reduction270

Redox reaction..... 270

Extraction of iron from iron ore..... 273

Topic 6 Organic chemistry.....279

Hydrocarbons..... 279

Alkanes..... 281

Alkenes..... 282

Biogas..... 285

End of term examinations292

Biology..... 292

Physics..... 297

Chemistry..... 303

Term

1

Biology



Topic number	Topic	Learning objectives
1	Structure and function of a cell	• Identify specialised cells • Draw and label specialised cells • State the functions of the specialised cells in relation to structure • Use a microscope to observe cell structure
2	Nutrition in plants	• State the factors that affect the rate of photosynthesis • Describe experiments to determine the factors, which affect the [rate] of photosynthesis • Describe the fate of the end products of photosynthesis • Testing for oxygen using a glowing splint • Identify the parts of the internal structure of a leaf • Describe how the leaf is adapted for photosynthesis
3	Nutrition in humans	• Draw and label parts of the alimentary canal and associated organs • Identify parts of the human alimentary canal • State the functions of parts of the human alimentary canal • Name types of teeth and their functions • Describe mechanical and chemical digestion • Explain the importance of digestion • Describe the function of a typical enzyme (amylase) • Identify the end products of digestion

4	Respiratory system	• State the differences between inhaled and exhaled air • Describe the role of the alveoli in gas exchange • Explain how the alveoli are adapted for gas exchange
5	Transport in plants	• Explain the process of transpiration • State the factors affecting the rate transpiration • Measure transpiration in a plant • Outline the importance of transpiration • Explain the terms plasmolysis and turgidity • Describe the effects of water loss and water gain in plant cells
6	Transport system in humans	• Describe the double circulatory system
7	Reproductive systems in plants	• Explain the structures of wind- and insect-pollinated flowers • Describe the process of germination • Investigate conditions necessary for germination • Calculate percentage germination
8	Reproduction in humans	• State the functions of the male and female reproductive systems • Describe the structure and functions of sex cells • Draw and label the sex cells • Describe the route of the sperm from the testis to the ovum after ovulation • Define fertilisation • Describe the menstrual cycle • Describe the role of the placenta • Identify the substances exchanged in the placenta

Topic 1 Structure and function of a cell

Learning objectives and activities

Learning objectives	Activities
Identify specialised cells	Drawing specialised cell structures
Draw and label specialised cells	Using a microscope or bioviewer to observe specialised cells
State the functions of the specialised cells in relation to structure	
Use a microscope to observe cell structure	

In Form 1 you learnt about the structure of plant and animal cells and that there are similarities and differences between them.

Specialised cells

Plants and animals are made up of many different types of cells. Not all cells have the same structure because different types of cells carry out different jobs or functions in the plant or animal.

There is a link between the structure of an object and its function. A pen has a specific structure so that it can be used for writing and a ruler has a specific structure so that you can use it to measure length. Just like a pen and a ruler, cells have a specific structure that enables them to carry out a specific function in an organism.

When cells have a specific structure to carry out a specific function they are **specialised**. Specialisation ensures that plants and animals can perform the life processes to stay alive and healthy. A group of cells with the same structure and function is called **tissue**.

Specialised cells in animals

Animals have many different types of specialised cells. The specialised cells in animals that you will learn about are red blood cells and muscle cells.

Red blood cells

In Form 1 you learnt that red blood cells are one of the components of blood and that their function is to transport oxygen to all the cells in the body. The structure of red blood cells makes them adapted to their function.

Word help

specialised cell: cells that are specialised are adapted in some way to their function

Red blood cells are filled with a red pigment called **haemoglobin** to enable them to transport oxygen. Haemoglobin combines with oxygen when blood passes through the lungs. It then releases this oxygen to all the cells in the body as it moves through the body in the circulatory system.

Mature red blood cells do not have a nucleus. This is to make space for the maximum number of haemoglobin molecules so that as much oxygen as possible can be transported by the red blood cells. Red blood cells are **bi-concave** discs. This means that the centre of the cell is thinner than the edges. Figure 1.1 shows the shape of a red blood cell. This shape gives the cells a larger surface area so that as much oxygen as possible can diffuse into the cells and combine with the haemoglobin.

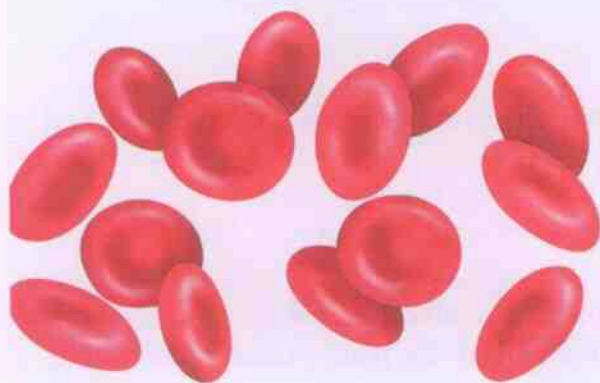


Figure 1.1 Red blood cells

Muscle cells

There are three types of muscle namely skeletal muscle, **cardiac muscle** and **smooth muscle**. Figure 1.2 shows the types of muscle cells.

Skeletal muscles are responsible for movement. These muscles are attached to bones. You will focus on how skeletal muscle cells are specialised.

The skeletal muscle cells are called muscle fibres because the cells are long and thin. The muscle cells have more than one nucleus per cell. When a cell has many nuclei it is called **multinucleate**. The muscle cells are very **elastic**. The specialised structure of muscle cells allows them to contract (become shorter) and relax to make bones move and so bring about the function of movement.

Something interesting

Red blood cells only live for 120 days. One red blood cell contains approximately 250 million molecules of haemoglobin!

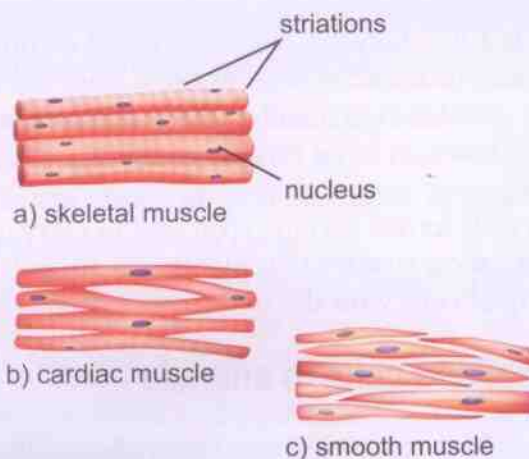


Figure 1.2 Types of muscle cells

Word help

haemoglobin: part of red blood cells that transport oxygen

bi-concave: concave on both sides

cardiac muscle: a type of muscle making up the walls of the heart

smooth muscle: muscle not under voluntary control, consisting of spindle-shaped cells

multinucleate: a cell containing at least two nuclei

elastic: ability to stretch and contract back to the original size or length

Activity 1.1 Practical task

Observing red blood cells and muscle cells

Work on your own

Aim: To observe the structure of red blood cells and skeletal muscle cells through a microscope

You will need:

light microscope
a prepared slide of red blood cells
a prepared slide of skeletal muscle cells

OR

bioviewer
bioviewer slides of red blood cells
bioviewer slide of skeletal muscle

Method

Follow the instructions.

1. Study the diagram of a microscope in Figure 1.3.
2. Follow the steps to set up the microscope. Do this every time you use a microscope.

- Switch the microscope on if it has a light switch. Turn the nosepiece so that the lowest power objective lens is "clicked" into position.
- Put the microscope slide on the stage and hold it in place using the stage clips.
- Using the coarse (large) focus knob, lower the objective lens down as far as it will go without touching the slide. Look from the side while doing this.
- Look through the eyepiece and adjust the diaphragm for the greatest amount of light. If there is a mirror, adjust the mirror.
- Slowly turn the coarse focus knob so that the objective lens moves up – until the image comes into focus. Use the fine focus knob for fine focus.
- Slowly move the stage with the stage adjustment knob so that the image is in the center of the field of view.

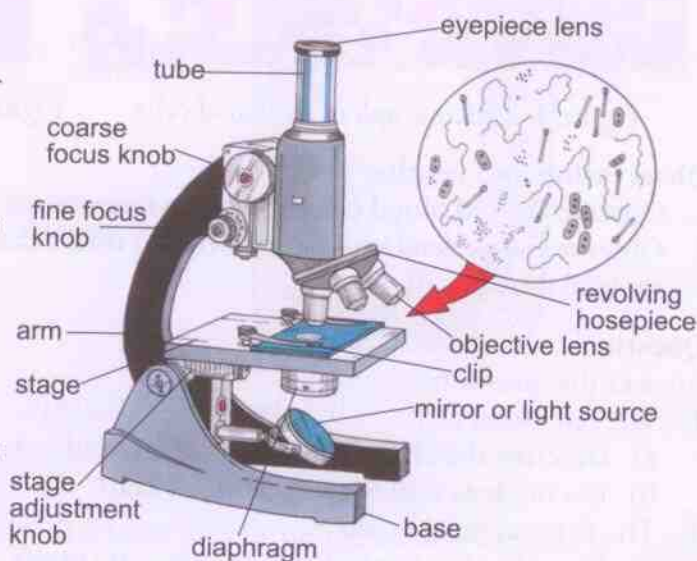


Figure 1.3A light microscope

Note that the image is upside down and backwards. When you move the slide to the right, the image goes to the left!

- Change to the next higher power objective lens. Focus using the fine adjustment knob. Do not allow the lens to touch the slide!
- Look through the eyepiece with one eye and keep the other eye open.
- When finished, raise the tube with the coarse focus knob and click the low power objective lens into position. Remove the slide. Switch off the microscope.

If microscopes or bioviewers are not available use the **micrographs** in Figure 1.4 and 1.5 for your observations.



Figure 1.4 Micrograph of red blood cells

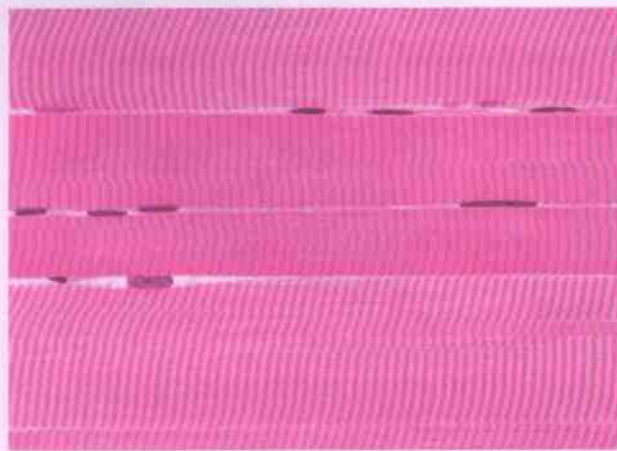


Figure 1.5 Micrograph of skeletal muscle cells

Observation and results

1. Observe the red blood cells under the microscope. Make a labelled drawing of a red blood cell.
2. Observe the skeletal muscle cells (fibre) under the microscope. Make a labelled drawing of a skeletal muscle cell.

Questions

Answer the questions.

1. The red blood cell
 - a) Describe the shape and colour of a red blood cell.
 - b) Is a nucleus visible in a red blood cell?
2. The skeletal muscle cell
 - a) Describe the shape of one muscle cell (fibre).
 - b) Is a nucleus visible?
 - c) How many nuclei can you see in one cell?

Activity 1.2

Work on your own

Answer the questions.

1. Compare a red blood cell and a skeletal muscle cell. Use a table.
2. Explain why red blood cells have a disc or bi-concave shape?

3. If there are many small cells compared to fewer larger cells, which do you think can absorb and carry more oxygen? Explain your answer.
4. Describe two ways in which the structure of a red blood cell is linked to its function.

Specialised cells in plants

The specialised cells in plants that you will learn about are palisade cells and root hair cells.

Palisade cells

In Form 2 you learnt that photosynthesis is a process that uses light, water and carbon dioxide to produce glucose in the presence of chlorophyll. The function of the palisade cells is photosynthesis. Palisade cells are found in a layer near the top of the leaf below the epidermis layer of cells. You will learn more about the internal structure of the leaf in Topic 2 Plant nutrition. Being at the top, near the surface of the leaf allows the palisade cells to absorb more light for photosynthesis. Figure 1.6 shows the structure of a palisade cell. The structure of palisade cells adapts them for photosynthesis. They are rectangular in shape so that as many as possible can fit long ways into the palisade layer of the leaf. Palisade cells contain many chloroplasts. **Chloroplasts** contain the green pigment chlorophyll that traps light for the process of photosynthesis.

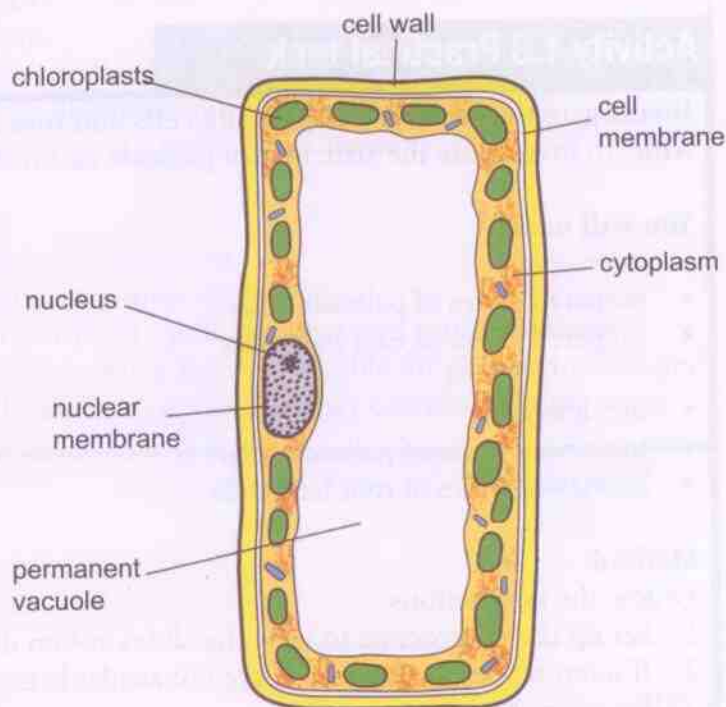


Figure 1.6 A palisade cell

Root hair cells

Roots consist of different types of specialised cells. The layer of cells on the outside of roots is called the epidermis layer. Some epidermis cells have long thin **extensions** called root hairs. These specialised epidermis cells are called root hair cells. In Form 1 you learnt how root hair cells in the roots of plants absorb water by osmosis.

Word help

chloroplasts: organelles in a cell containing chlorophyll and site for photosynthesis
extensions: an addition or add-on to elongate something
micrograph: a photo taken through a microscope

The function of root hair cells is to absorb water from the soil into the roots. The long, thin extensions on root hair cells increase the surface area of the root and cause them to have more contact with the soil and the soil water. Because of the large surface area, the root hair cells can absorb as much water and dissolved minerals as possible. Figure 1.7 shows the structure of root hair cells.

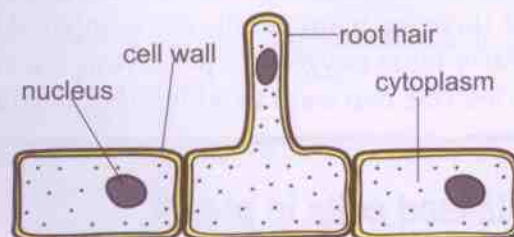


Figure 1.7 A root hair cell

Activity 1.3 Practical task

Investigate the structure of palisade cells and root hair cells.

Work in pairs

Aim: To investigate the structure of palisade and root hair cells through a microscope

You will need:

- microscope
- prepared slides of palisade cells
- prepared slides of root hair cells

OR

- bioviewer
- bioviewer slides of palisade cells
- bioviewer slides of root hair cells

Method:

Follow the instructions.

1. Set up the microscope to view the slides as you did in Activity 1.1.
2. If microscopes or bioviewers are not available use the micrographs in Figure 1.8 and Figure 1.9 for your observations.



Figure 1.8 A micrograph of palisade cells

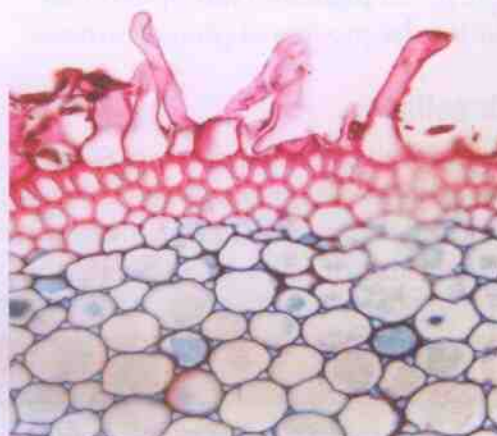


Figure 1.9 A micrograph of root hair cells

Observation and results

1. Observe the slide with the palisade cells through the microscope. Make a labelled drawing of one cell.
2. View the slide with the root hair cells through the microscope. Make a labelled drawing of one cell.

Questions

Answer the questions

1. a) Where are palisade cells found in the plant?
b) Where are root hair cells found in the plant?
2. Do root hair cells have a similar shape to palisade cells? Explain your answer.
3. How is the internal structure of a root hair cell different from a palisade cell?
4. Which are more numerous – root hair cells or roots? Explain your answer.

Summary

Different cells are specialised to carry out their specific functions:

- Red blood cells lack a nucleus so that they can carry more oxygen.
- Muscle cells are long and elastic so they can contract to pull on bones and cause movement.
- Palisade cells have many chloroplasts to absorb as much light as possible for photosynthesis.
- Root hair cells have extensions to increase the surface area so that they can absorb more water.

Topic assessment

Answer the questions.

1. a) Explain the function of red blood cells? [1]
b) How does the structure of a red blood cell help it to perform its function? [2]
c) Make a drawing of a red blood cell to show its shape. [2]
5. a) Muscle fibres are elastic. How does this help their function? [2]
b) Make a labelled drawing of a muscle fibre. [2]
6. a) What numerous structures inside palisade cells allow them to trap as much light as possible for photosynthesis? [1]
b) Do a labelled drawing of a palisade cell. [4]
7. a) Describe the function of root hair cells? [2]
b) Suggest how the root hair cell structure helps the cells to perform their function. [2]
c) Complete a labelled drawing of root hair cell. [4]

Total marks [20]

Topic 2 Nutrition in plants

Learning objectives and activities

Learning objectives	Activities
• State the factors that affect the rate of photosynthesis	
• Describe experiments to determine the factors, which affect the [rate] of photosynthesis	• Investigating factors affecting the rate of photosynthesis
• Describe the fate of the end products of photosynthesis	• Testing for starch in a leaf • Testing for oxygen using a glowing splint
• Identify the parts of the internal structure of a leaf	• Observing the internal leaf structure using a bioviewer or microscope
• Describe how the leaf is adapted for photosynthesis	• Examining the external leaf structure

In Form 2 you learnt that plants are producers because they produce food (carbohydrates) by the process of photosynthesis. These carbohydrates are consumed by animals. You also learnt the word equation for photosynthesis, which represents the process of photosynthesis.

In Form 2 you learnt that photosynthesis in plants takes place in organelles called chloroplasts containing chlorophyll.

In Topic 1 you learnt about specialised cells in plants namely palisade cells that contain chloroplasts and are adapted for photosynthesis.

Photosynthesis

Photosynthesis is the process by which green plants make food. During the process of photosynthesis sunlight energy is converted into chemical energy. During photosynthesis the plants use water from the soil and carbon dioxide from the air and energy from sunlight to make food (glucose) in the presence of chlorophyll and release oxygen into the air. Oxygen is released as a **by-product**. There are various **factors** that affect the **rate** of photosynthesis. From the word equation of photosynthesis one can identify the factors affecting the process of photosynthesis and the products that form.

Word help

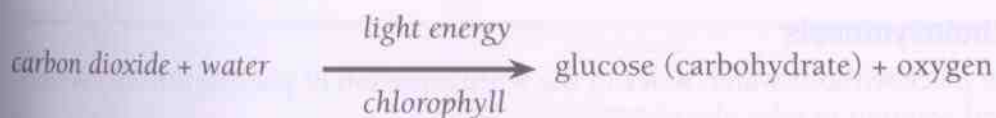
photosynthesis: the process by which green plants use light energy to manufacture food (carbohydrate) using carbon dioxide and water and releasing oxygen as a by-product

rate: the speed (fast or slow) at which something happens

by-product: not the main or primary product

factor: one of the things that causes something to happen or change

intensity: how bright the light is



For the process of photosynthesis plants need water, carbon dioxide, light and the green pigment chlorophyll in their leaves. If one of these factors is absent or only available at a low level, the process will slow down or stop. Fewer carbohydrates will be produced and less oxygen released. The process of photosynthesis is summarised in Figure 2.1.

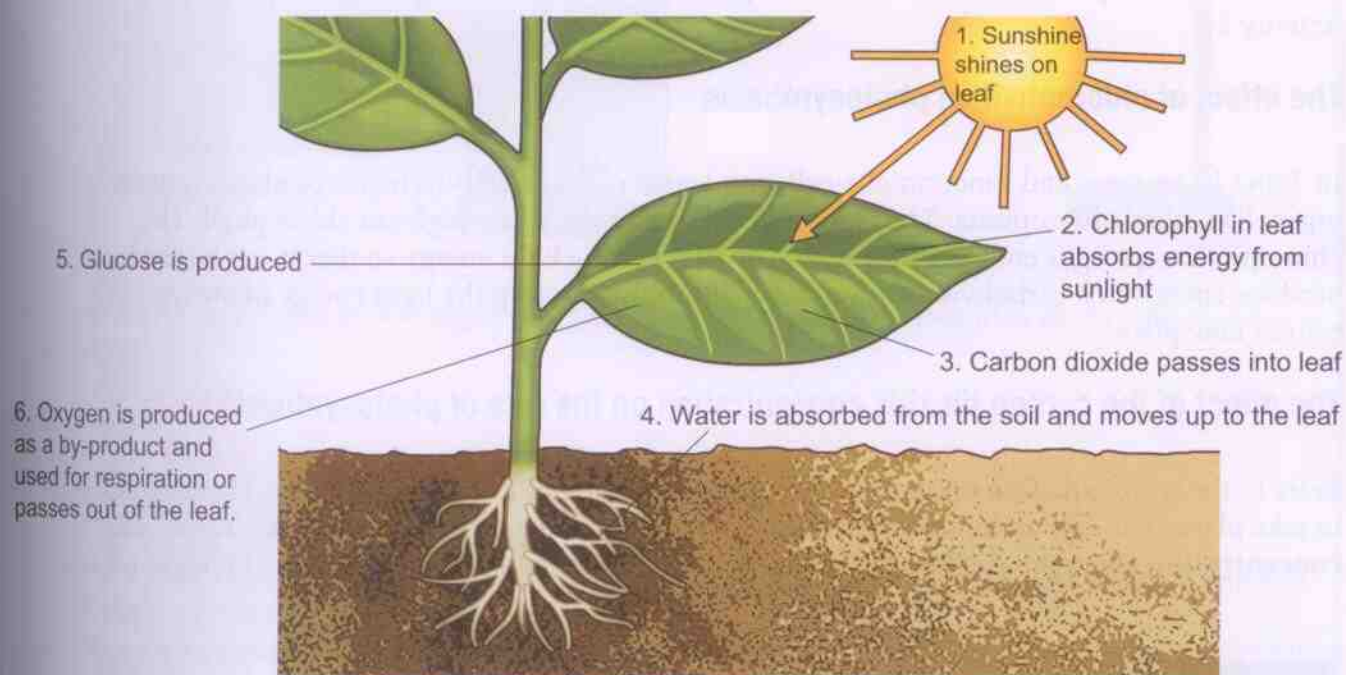


Figure 2.1 Summary of the process of photosynthesis

1. Sunshine shines on leaf
2. Chlorophyll in leaf absorbs energy from sunlight
3. Carbon dioxide passes into leaf
4. Water is absorbed from the soil and moves up to the leaf
5. Glucose is produced
6. Oxygen is produced as a by-product and used for respiration or passes out of the leaf.

Factors affecting photosynthesis

It is possible to investigate in a school laboratory how some factors affect photosynthesis. We will investigate how carbon dioxide concentration and light **intensity** affect the rate of photosynthesis by using a group of water plants called pond weed. The rate of photosynthesis can be determined by counting the number of oxygen bubbles released by the pond weed in a minute. The amount of water and chlorophyll available in a leaf will also affect photosynthesis. These investigations are more complicated and you will not investigate these factors yourself.

The effect of water on photosynthesis

Photosynthesis cannot take place without water. Refer to the word equation of photosynthesis. Water is necessary for the chemical reaction to take place.

Plants also need water to stay turgid. You will learn more about turgidity in Topic 5 Transport systems. If a plant does not have enough water the stomata will close. The stomata are openings on the bottom surface of the leaf through which the plant absorbs carbon dioxide. The availability of carbon dioxide affects the rate of photosynthesis. We investigate the effect of carbon dioxide on photosynthesis in Activity 2.1.

The effect of chlorophyll on photosynthesis

In Topic 1 Structure and function of a cell, you learnt palisade cells in leaves of plants contain organelles called chloroplasts. The chloroplasts contain the green pigment chlorophyll. The chlorophyll traps light energy and converts or changes the light energy so that it can be used to produce energy-rich carbohydrates. Without chlorophyll to trap the light energy photosynthesis cannot take place.

The effect of the carbon dioxide concentration on the rate of photosynthesis

Refer to the word equation of photosynthesis. Carbon dioxide is necessary for the chemical reaction to take place. The carbon in carbon dioxide is needed to make carbohydrates like glucose. The **concentration** of carbon dioxide in the air surrounding the plant affects the rate of photosynthesis.

Activity 2.1 Practical task

The effect of carbon dioxide concentration on the rate of photosynthesis Work in groups

Aim: To investigate how the carbon dioxide concentration affects the rate of photosynthesis

You will need:

- 600 cm³ beaker
- 500 ml water
- test tube
- spatula
- electronic balance
- 1 metre ruler or measuring tape
- paper clip
- scissors
- stop watch or timer
- forceps
- funnel
- plasticine
- bright light source
- a piece of pond weed e.g. Elodea
- sodium bicarbonate (bicarbonate of soda, baking soda)

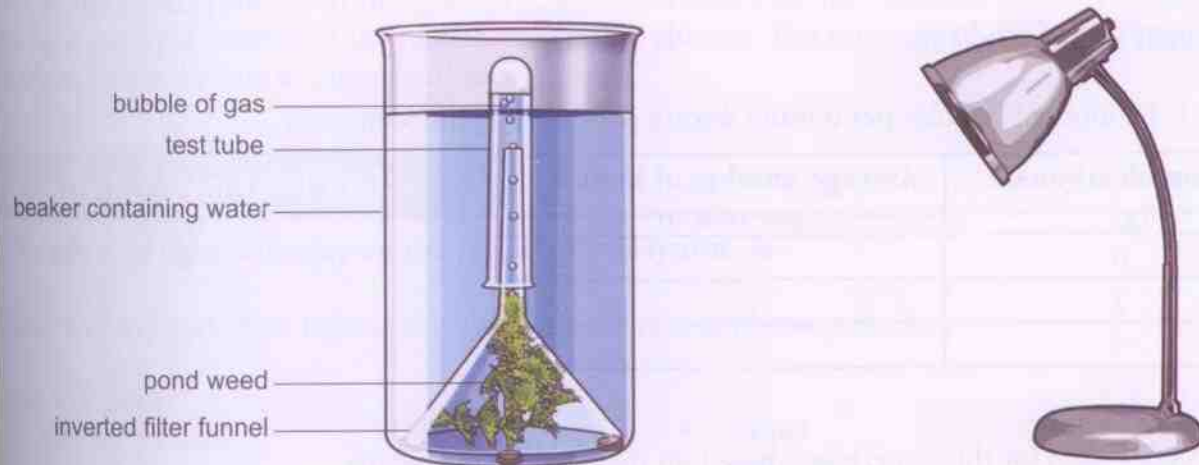


Figure 2.2 An experiment to investigate the effect of carbon dioxide concentration on the rate of photosynthesis

Method

Follow the instructions.

1. Pour 500 ml water in the beaker. (See Figure 2.2.)
2. Cut a piece of pond weed (5–10 cm) and gently crush its stem. Place the paper clip on the end of the stem to weigh it down and place it in the beaker.
3. Place an inverted (upside-down) funnel over the pond weed and hold it above the bottom of the beaker using the plasticine. The funnel must be raised above the bottom of the beaker to allow dissolved gases to diffuse freely through the water in the beaker.
4. Fill a test tube with water and cover the open end with your finger while you turn it upside down. Gently place the test tube over the stem of the filter funnel and lower it until it sits on top of the filter funnel as shown in Figure 2.2.
5. Place the light 50 cm away from the beaker.
6. Look for bubbles coming from the cut end of the pondweed. The bubbles are an indication of the rate of photosynthesis. The more bubbles produced the higher the rate of photosynthesis.
7. Count the number of bubbles produced in one minute. Repeat three times and calculate the average number of bubbles per minute. Record the average number in a table. See Table 2.1.
8. Add 2 g sodium bicarbonate to the water in the beaker using a spatula. Stir the water to dissolve the sodium bicarbonate. Sodium bicarbonate dissolves in water to increase the carbon dioxide concentration.
9. Wait for two minutes to allow the experiment to adjust to the new carbon dioxide concentration. Count the number of bubbles produced in one minute. Repeat three times and calculate the average number of bubbles per minute. Record the average number in a table (see Table 2.1).
10. Add another 2 g sodium bicarbonate and stir. Repeat step 9.

Observations and results

Record your results in a table.

Table 2.1: Number of bubbles per minute during pond-weed photosynthesis

Sodium bicarbonate (g)	Average number of bubbles per minute
0	
2	
4	

Conclusion

Write a conclusion for this experiment based on the aim and the results.

Questions

Answer the questions.

- a) What chemical releases carbon dioxide when dissolved in water?
b) Explain why this chemical is used in this experiment?
- a) Name two factors that must be kept constant during the investigation.
b) Why must these factors be kept constant?
- Suggest why the rate of photosynthesis becomes constant.

The effect of carbon dioxide concentration on the rate of photosynthesis

As the concentration of carbon dioxide in the air or water surrounding a plant increases, the rate of photosynthesis also increases. The rate of photosynthesis increases to a maximum level. After this point the rate remains the same (constant). It reaches a maximum because the light intensity may be too low or photosynthesis in the plant just cannot take place any faster. The effect of carbon dioxide concentration on the rate of photosynthesis can be represented by a line graph. See Figure 2.3.

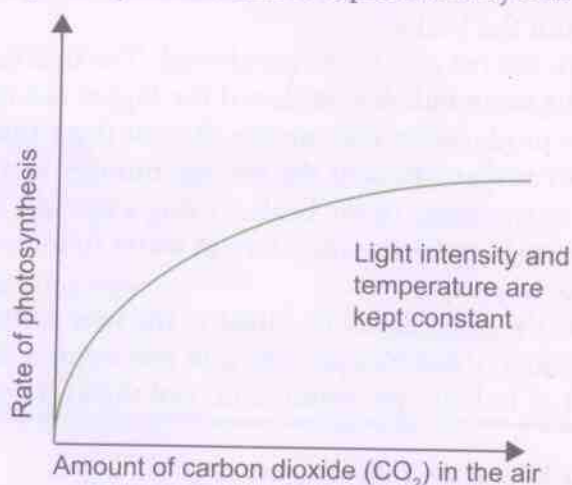


Figure 2.3 How the amount of carbon dioxide affects the rate of photosynthesis

The effect of light on the rate of photosynthesis

Refer to the photosynthesis word equation. Light is necessary for the chemical reaction to take place. The light energy is trapped in the carbohydrates like glucose. The intensity of the light surrounding the plant affects the rate of photosynthesis.

Activity 2.2 Practical task

The effect of light intensity on the rate of photosynthesis

Aim: To investigate how light intensity affects the rate of photosynthesis

You will need:

- 600 cm³ beaker
- 500 ml water
- test tube
- spatula
- electronic balance
- 1 metre ruler or measuring tape
- paper clip
- scissors
- stop watch or timer
- forceps
- funnel
- plasticine
- bright light source
- heat shield (piece of glass or another beaker with water)
- thermometer
- a piece of pond weed e.g. Elodea
- sodium bicarbonate (bicarbonate of soda, baking soda)

Method

Follow the instructions.

1. Set up the apparatus as in Activity 2.1. Dissolve 4 g of sodium bicarbonate in the water before you add it to the beaker.
2. Check the temperature in the beaker with the thermometer from time to time to make sure that it remains constant throughout the experiment. If the temperature rises place the heat shield between the lamp and the experiment. Replace the sodium bicarbonate solution if necessary.
3. Place a lamp 20 cm away from the beaker. Allow the plant to adjust to the light intensity for two to three minutes.
4. When the air bubbles are produced at a regular rate, count the number of bubbles produced in one minute. Repeat three times. Calculate an average and record the average number of bubbles in a table.
5. Repeat step 3 but place the lamp with increasing distances from the experiment. Repeat at 30 cm, 40 cm and 50 cm. As the distance between the plant and the lamp increases the light intensity decreases.
6. Place the beaker with the pond weed, funnel and test tube on a sunny windowsill. Leave it there for a couple of days. It will be used in Activity 2.4 to test whether the gas forming the bubbles is oxygen.

Observation and results

Record your results in a table.

Table 2.2: How the intensity of light affects photosynthesis

Distance between plant and light source	Average number of bubbles per minute
20	
30	
40	
50	

Conclusion

Write a conclusion for this experiment based on the aim and the results.

Questions:

Answer the questions

- a) Why is the plant kept in a solution of sodium hydrogen carbonate?
b) Suggest a reason why the temperature of the sodium bicarbonate solution needs to be kept constant.
- Identify any source of error or inaccuracy in this experiment.
- Use the data below to draw a line graph showing the effect of light intensity on the rate of photosynthesis

Hint: Remember that the closer the lamp the higher the light intensity.

Table 2.3: How the intensity of light affects photosynthesis

Distance between plant and light (cm)	Average number of bubbles per minute
10	80
20	80
30	60
40	35
50	10

The effect of light on the rate of photosynthesis

Light is needed to supply the energy for photosynthesis so that energy-rich carbohydrates like glucose can be produced. As the light intensity increases, the rate of photosynthesis also increases. The rate of photosynthesis increases to a maximum level. After this point the rate remains the same (constant). It reaches a maximum because the carbon dioxide concentration may be too low or photosynthesis in the plant just cannot take place any faster. The effect of light intensity on the rate of photosynthesis can be represented by a line graph. See Figure 2.4.

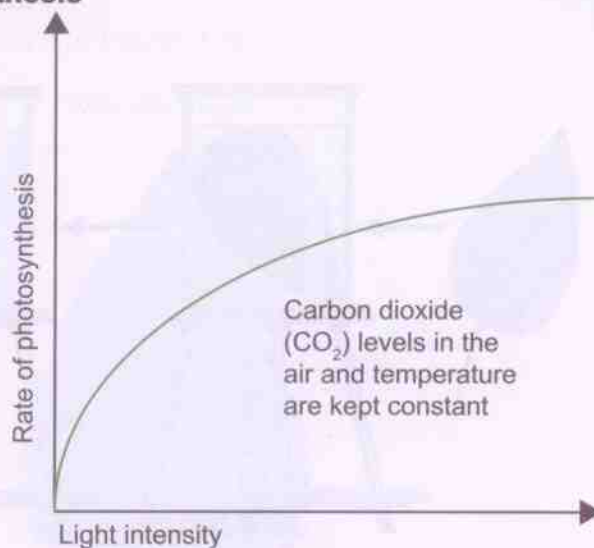


Figure 2.4 Graph showing the effect of light intensity on the rate of photosynthesis

The products formed by photosynthesis

Plants use light, carbon dioxide and water to produce products carbohydrates like glucose and oxygen. The glucose produced by photosynthesis can be used for the process of respiration or stored in the form of starch. The presence of starch in a leaf is an indication that photosynthesis took place. We can also do an experiment to show that the gas produced during photosynthesis is oxygen. You observed the test for oxygen in Form 1.

Activity 2.3 Practical task

Testing a leaf for starch Work in pairs

Aim: To test for the presence of starch in a leaf.

You will need:

- methylated sprits
- iodine solution
- beaker containing water
- tripod and gauze
- Bunsen burner
- test tube
- forceps
- white tile
- dropper
- a green leaf

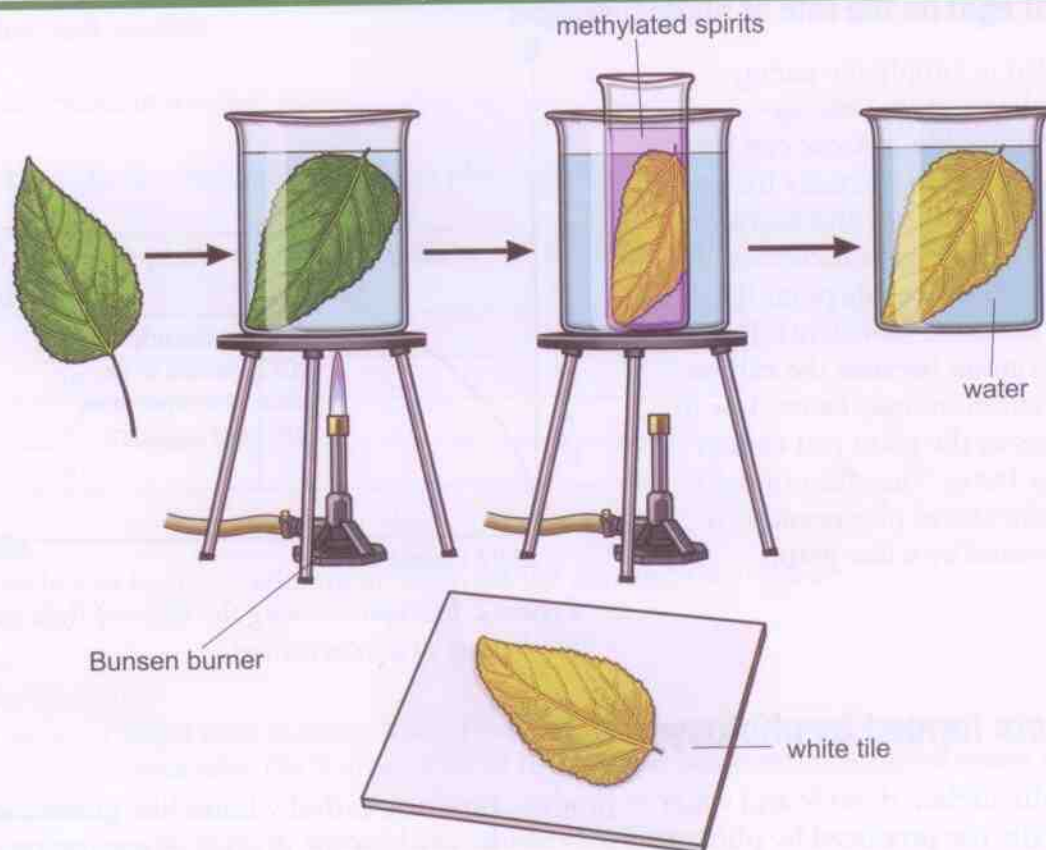


Figure 2.5 Testing a leaf for starch

Method

Follow the instructions. See Figure 2.5.

1. Remove a leaf from a healthy plant that is growing outside. Place the leaf in a beaker of boiling water. This kills the cells of the leaf so the iodine solution will be able to get to the starch inside the cells.
2. Use the forceps to move the leaf from the beaker of water into the test tube containing methylated spirits. **TURN OFF THE BUNSEN BURNER.** Now place the test tube in the beaker of hot water. The methylated spirits will boil as it has a lower boiling point than water. The green chlorophyll moves into the methylated spirits from the leaf. Continue boiling the leaf until all the chlorophyll has been removed.
3. Use the forceps to move the leaf from the methylated spirits back into the hot water in the beaker. Wash the leaf in the water.
4. Place the leaf on the white tile and spread it out so that it is flat. Use the dropper to drop iodine solution over the surface of the leaf.
5. Observe the colour of the leaf. A blue black colour indicates the presence of starch. The presence of starch in turn indicates that photosynthesis has taken place.

Observations and results

A leaf that has tested positive for starch is shown in Figure 2.6b).

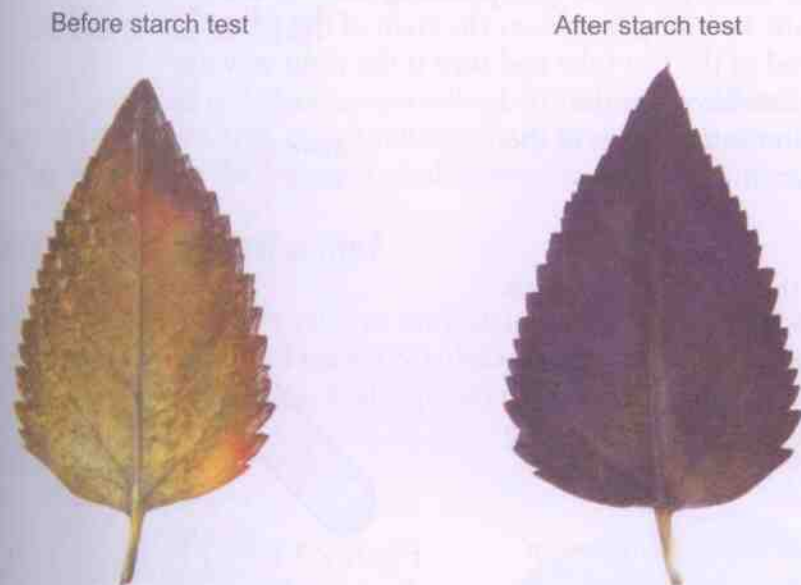


Figure 2.6 A leaf containing starch a) before the starch test b) after the starch test

Conclusion

Based on the aim and the results, write a conclusion for this experiment

Questions

Answer the questions.

1. Why is the leaf boiled in water?
2. What happens when the leaf is boiled in alcohol?
3. Describe the safety precaution that should be taken. Explain why.
4. If the leaf turns blue-black at the end of the starch test what does this mean?
5. Sometimes the leaf does not turn blue-black (or the iodine stays the same brown colour) at the end of the starch test. Suggest what this might mean.

Activity 2.4 Practical task

Investigating whether oxygen is produced during photosynthesis Work as a class

Aim: To investigate whether oxygen is produced during photosynthesis

You will need:

- the experiment set up in Activity 2.2
- matches or a lighter
- a wooden splint

Method

Follow the instructions.

1. Remove from the windowsill the experiment set up during Activity 2.2.
2. Carefully remove the test tube by lifting it up over the stem of the filter funnel.
3. Place your finger over the end of the test tube and turn it the right way up.
4. Light a wooden splint and then blow out the flame. Place the glowing splint into the top of the test tube after removing your finger.

Observations and results

Describe what happened when the glowing splint was inserted into the test tube.

1. Does the splint glow brighter or ignite?
2. Does the splint not burn?
3. If it does ignite, what do you think this indicates?

Conclusion

Based on the aim and the results, write a conclusion for this experiment.

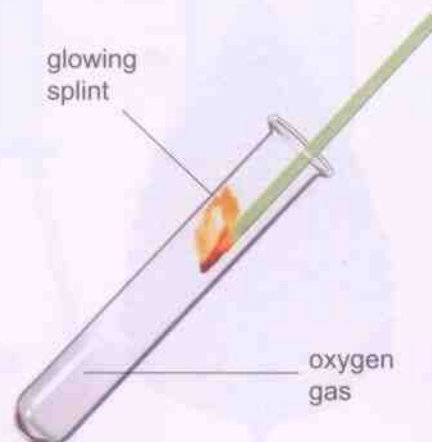
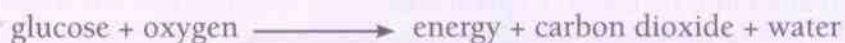


Figure 2.7 Testing for oxygen using a glowing splint

The fate of the end products of photosynthesis

During the process of photosynthesis, plants produce the products glucose and oxygen. Glucose is the main product and oxygen is a byproduct. Both these products are needed by the plant to stay alive. All the cells in the plant require glucose and oxygen for respiration. Respiration is a process that takes place in all living cells and during this process energy is released for the cell to use. In Form 2 you learnt about respiration and the word equation for respiration. The word equation for respiration is:



The energy produced by respiration is used by the plant for growth, reproduction and other life processes. You will learn more about respiration in Form 4.

The glucose (a complex sugar) produced by photosynthesis in the leaves needs to be transported to the other plant cells where it is required. The glucose is converted to another carbohydrate called sucrose (a simple sugar). The sucrose is transported to the root and other parts of the plant by tissue called phloem. The transport of sucrose in **phloem** is called **translocation**. Some of the glucose that is not immediately used for respiration can be stored in the plant form of starch. Some plants have special **organs** that store starch. A potato is an example of a special stem that stores starch.

Some of the glucose produced by photosynthesis is moved to growing tissue, for example the tips of the roots and shoots. The glucose is converted into cellulose to form cell walls for new cells. The glucose can also be used to form proteins and fats and oils (lipids). These are used to form more plant organs, for example leaves, flowers and seeds.

Some of the oxygen produced by photosynthesis is used for respiration in the plant cells. The rest of the oxygen leaves the plant through the stomata into the air.

The structure of a leaf

In Topic 1 you learnt that cells have a specialised structure that adapts them to perform a specific function. Just like the structure of cells allows them to perform a specific function, the structure of plant leaves is adapted for the function of photosynthesis.

The external structure of a leaf

The **external** structure of leaves refer to the outside, visible parts of the leaf. Most leaves are covered by waxy cuticle on the outside. This waxy substance is waterproof and prevents the plant from losing too much water through its leaves. It also protects the leaf.

On the surface of most leaves you will see veins. The largest main vein is called the midrib. The veins transport substances to and from the leaf. Leaves are attached to the stem by a petiole. The petiole can change the position of the leaf so that it always receives as much sunlight as possible for photosynthesis. Leaves come in many different sizes and shapes, but many leaves are thin and flat with a large surface area. The flat part of the leaf is called the lamina and the edge is called the margin.

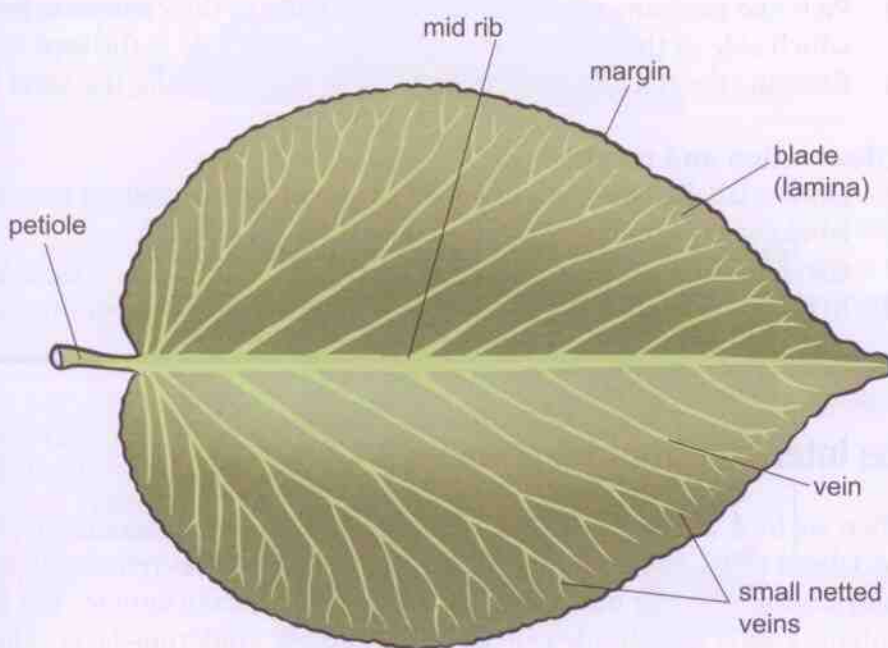


Figure 2.8 The external structure of a leaf

Word help

external: the outside

translocation: the transportation of food molecules from one part of a plant to another in the phloem

phloem: the vascular tissue in plants which transport sugars downward from the leaves

organs: a part of an organism that performs a particular function

Activity 2.5 Practical task

Examining the external structure of a leaf Work on your own

Aim: To examine the external structure of a leaf

You will need:

- a variety of leaves
- a hand lens

Method

Follow the instructions.

1. Go into the school yard and observe the different sizes and shapes of the leaves that you see.
2. Note how the leaves are orientated (their position) in relation to where the sun is situated.
3. Pick one medium-sized leaf and take it back to the classroom for closer examination. Note which side of the leaf was facing the sun. This side is the top.
4. Examine the top and bottom surfaces of the leaf using the hand lens.

Observation and results

1. Draw a labelled diagram of your leaf showing the external structures. You can use Figure 2.8 to label your diagram.
2. Give your diagram a suitable heading.
3. In a table list two differences between the top and the bottom surfaces of the leaf.

The internal structure of leaves

When we look at a cross-section of a leaf we will see that it consists of layers of different types of specialised cells. The structure of the inside of the leaf is referred to as its **internal** structure.

Study Figure 2.9. The outer layer of the leaf is the waxy cuticle. You learnt in Topic 1 that the leaf contains a layer of palisade cells below the upper epidermis layer. The layer of palisade cells is called the palisade mesophyll. Below the palisade mesophyll layer there is a layer of spongy mesophyll cells. These cells are round in shape with many air spaces between the cells. The palisade mesophyll cells and the spongy mesophyll cells contain chloroplasts. All cells containing chloroplasts are called mesophyll cells.

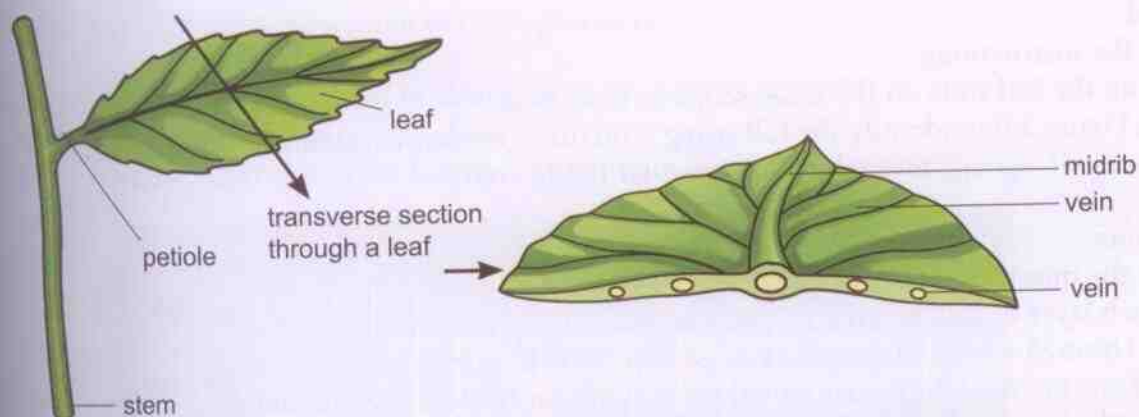
Word help

internal: the inside

intercellular spaces: refers to spaces between cells **internal:** the inside

The lower epidermis layer contains many pores (openings) called stomata. These pores are surrounded by specialised epidermal cells called guard cells. The guard cells can open or close the stomata (singular is stoma).

A vein is found inside the leaf to transport substances to and from the leaf. The vein consists of specialised tissue called xylem and phloem. Xylem transports water, necessary for photosynthesis, to the leaf. The phloem transports sucrose to other parts of the plant by translocation. Together the xylem and phloem make up the vascular tissue.



Leaf adaptation for photosynthesis

- The leaf structure is adapted to perform its function of photosynthesis.
- Surface area: the surface area of the leaf is large to absorb as much light as possible.
- Cuticle: transparent to allow light to enter the leaf for photosynthesis.
- Epidermis: thin, transparent layer of cells to allow light through for photosynthesis.
- Palisade mesophyll: packed tightly at the top of the leaf; many chloroplasts to absorb light for photosynthesis.
- Stomata: openings that allow carbon dioxide to enter the leaf from the air for photosynthesis.
- Vascular tissue: transport water in (from the roots) and sucrose out of the leaf.
- Spongy mesophyll: loosely packed cells with intercellular spaces. The cells contain chloroplasts for photosynthesis.
- Air spaces: allow for the easy movement of gases oxygen, carbon dioxide and water vapour.

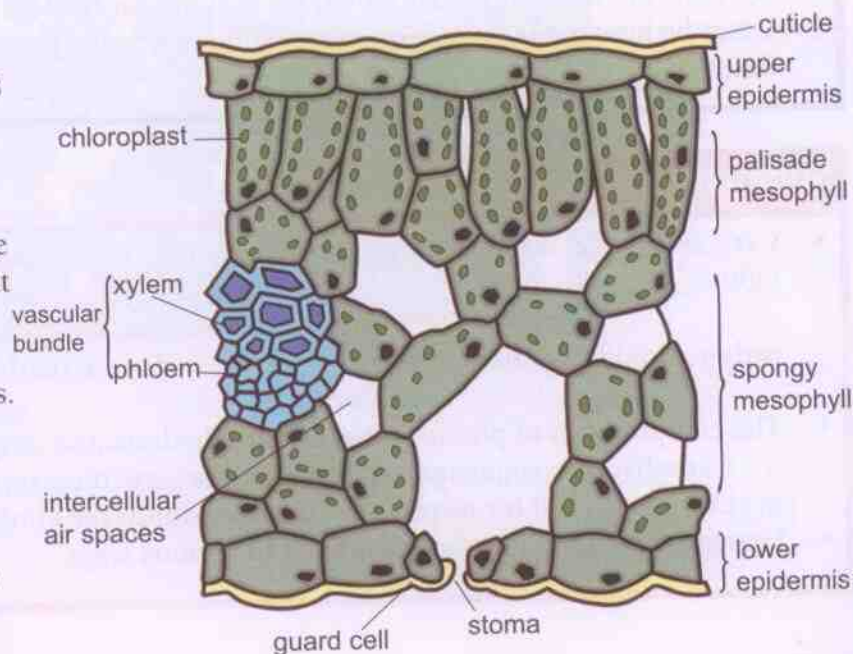


Figure 2.9 Cross-section through a leaf showing the internal structure of a leaf

Activity 2.6 Practical task

Observing the internal structure of a leaf

Work on your own

You will need:

- microscope
- slide of a leaf section
- OR a bioviewer and slide of a leaf section

Method

Follow the instructions.

1. Set up the leaf slide on the microscope or bioviewer.
2. Use Figure 2.9 to identify the following structures: epidermis, stomata, guard cells, palisade mesophyll, spongy mesophyll and vascular tissue (vein).

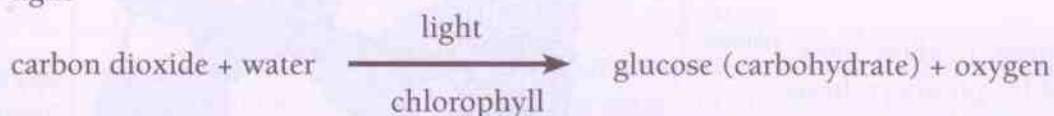
Questions

Answer the questions

1. Which layer of cells has the most chloroplasts for photosynthesis?
2. a) Through which structures does carbon dioxide enter a leaf?
b) How many of these structure do you count on the leaf on your slide?
3. Describe how the leaf that you are viewing is adapted for the process of photosynthesis. Only describe adaptations that you can see.

Summary

- Carbon dioxide, water, light and chlorophyll are factors which affect photosynthesis.



- The end products of photosynthesis (carbohydrate and oxygen) are used:
 - o Carbohydrate is transported and used for structures or stored.
 - o Oxygen is used for respiration or released into the atmosphere.
- Leaves are adapted for photosynthesis in various ways.

Topic 2 Assessment

Answer the questions.

1. a) State two factors that affect the rate of photosynthesis. [2]
b) Which other two factors do plants need for the process of photosynthesis? [2]
2. List the steps needed to test a leaf for starch. [4]
3. a) Name the two products of photosynthesis. [2]
b) Describe what happens to each of these products. [2]
4. a) How are most leaves adapted to absorb as much light as possible? [1]
5. Look at Figure 2.10. It shows the internal structure of a leaf.

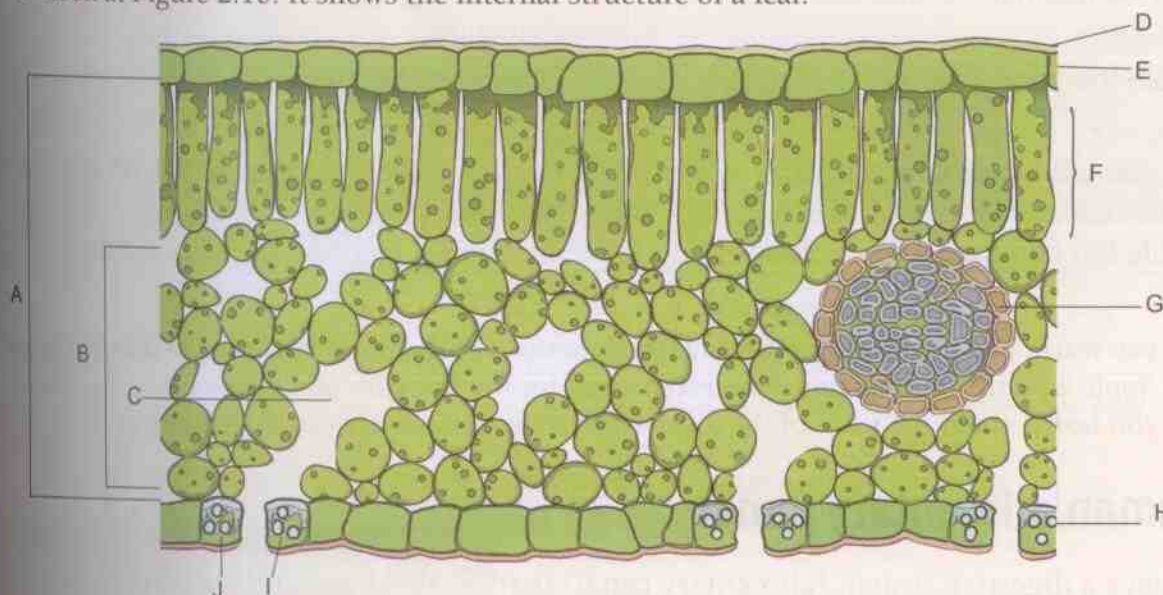


Figure 2.10 The internal structure of a leaf.

Name B to E Next to each named part suggest how the structure helps to adapt the leaf for photosynthesis. [12]

Total marks [25]

Safety

Safety box (Practical tasks)

Care should be taken to keep the alcohol away from the flame as alcohol is highly flammable.

Always boil the alcohol in a beaker with boiling water away from the flame.

Check that the Bunsen is turned off correctly.

Take great care with the glowing splint - it can burn you or set items alight.

Topic 3 Nutrition in humans

Learning objectives and activities

Learning objectives	Activities
• Draw and label parts of the alimentary canal and associated organs	• Watching a video simulation to examine the human alimentary canal
• Identify parts of the human alimentary canal	• Identifying the parts of the human alimentary canal using a chart
• State the functions of parts of the human alimentary canal	
• Name types of teeth and their functions	• Discussing the structure and functions of teeth • Observing types of teeth on a skeleton model
• Describe mechanical and chemical digestion	• Discussing mechanical and chemical digestion
• Explain the importance of digestion	• Demonstrating increase in surface area when a solid is broken into smaller pieces
• Describe the function of a typical enzyme (amylase)	• Experimenting using enzymes on different food substrates
• Identify the end products of digestion	• Testing for glucose, proteins and fats

In Form 1 you learnt about a balanced diet and the functions of the nutrients found in different kinds of food. In Topic 2 Nutrition in plants, you tested a leaf for the presence of starch. In Form 2 you learnt about the parts of the digestive system (alimentary canal) in humans.

The human alimentary canal

Humans have a digestive system (alimentary canal) that breaks down and converts the food we eat into a form that the body can absorb and use.

The alimentary canal in humans is a long specialised tube. Each part is specialised for a specific function. Study Figure 3.1 to identify the main parts of the alimentary canal and the functions of each part.

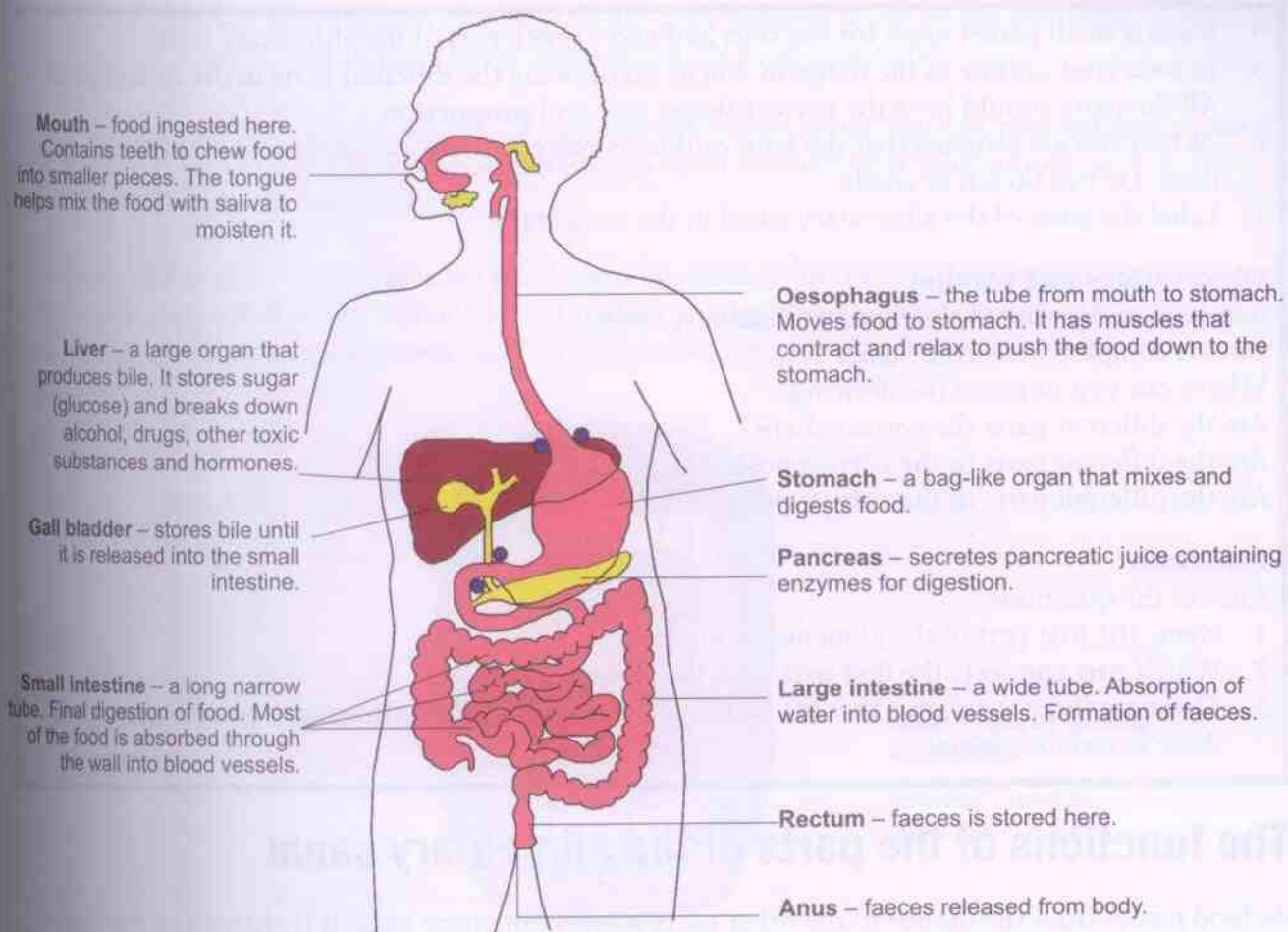


Figure 3.1 The human alimentary canal

Activity 3.1 Practical task

Drawing and labelling the parts of the alimentary canal

Work on your own

Aim: To draw the human alimentary canal

You will need:

- a sharp HB pencil
- a sheet of A4 paper
- a chart of the human alimentary canal

Method

Follow the instructions.

1. Use a chart of the human alimentary canal or Figure 3.1 The human alimentary canal.
2. Measure the length and width of the diagram.
3. Mark these points on your page.

4. Make a small pencil mark for the start and end of each part of the alimentary canal.
5. Draw a faint outline of the diagram. Focus on drawing the different parts in the correct position. All the parts should have the correct shape, size and **proportion**.
6. When you are satisfied that the faint outline is correct, draw the lines more clearly using solid lines. Do not sketch or shade.
7. Label the parts of the alimentary canal in the drawing.

Observations and results

Assess your drawing of the human alimentary canal.

Does it compare well with Figure 3.1?

Where can you improve the drawing?

Are the different parts the correct shape?

Are the different parts in the correct position?

Are the different parts in the correct proportion?

Questions

Answer the questions.

1. Name the first part of the alimentary canal.
2. Which part connects the first part with the stomach?
3. There are three structures that are NOT part of the tube of the alimentary canal. Name two of these associated organs.

The functions of the parts of the alimentary canal

As food passes from the mouth to the other parts of the alimentary canal it is changed so that the body can use the nutrients in the food. The food we eat undergoes five processes namely ingestion, digestion, absorption, assimilation and egestion. The alimentary canal is adapted to perform these processes or functions.

Ingestion

Ingestion is the process of taking in food through the mouth (eating) and chewing it into smaller pieces. We put food into the alimentary canal through the mouth. The mouth is the opening to the digestive system. After food is put into the mouth it is chewed by the teeth and swallowed.

Human teeth

Human teeth, through the act of chewing are responsible for the breaking down of large pieces of food into smaller pieces. There are four types of teeth in the mouth. Each type of tooth is adapted to its function.

Word help

ingestion: the process of taking in food

proportion: the correct or appropriate relationship between the size, shape, and position of the different parts of something

Incisors: the flat teeth with the sharp edges. You use your incisors to cut or bite off pieces of food. An adult has eight incisors, four in their lower jaw and four in their upper jaw.

Canines: found on either side of the incisors and are the pointy teeth. You use your canines to tear food and sometimes to grip the food that you eat. An adult normally has four canines, two in the upper jaw and two in the lower jaw.

Premolars and molars: the larger teeth and have a flat surface on the top. The top surface contains a number of bumps called cusps. Premolars and molars are the teeth used for chewing and grinding our food. Premolars are found next to our canines and molars are at the back of our mouth.

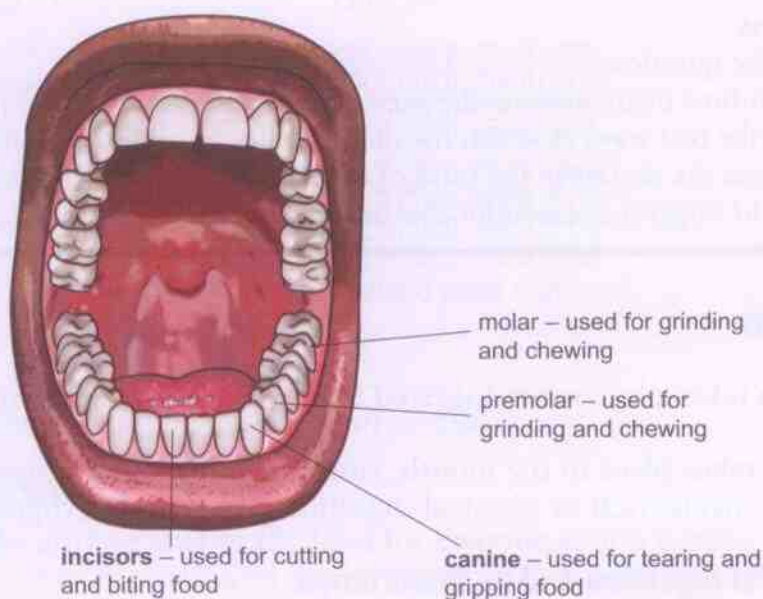


Figure 3.2 Human teeth types and their function

Activity 3.2 Practical task

Observing and examining human teeth types

Work with a partner

Aim: To observe and examine the structure and functions of human teeth

You will need:

a skeleton model of a human

Method

1. Examine the teeth in the jaw of the skeleton model.
2. Identify the four different types of teeth.
3. Count the number of each type of teeth.

	Learner A	Learner B
1	Ask your partner which teeth are used for eating tough, hard food. Why these teeth?	
2		Ask your partner why we have canines.
3	Ask your partner what incisors are used for. Why?	
4	For each type of tooth suggest the kinds of foods they are used for.	

Questions

Answer the questions

1. Count how many incisors the skeleton has and describe their position in the mouth.
2. Describe two ways in which the molars differ from the incisors.
3. How are the molars in the front of the mouth different from the molars in the back of the mouth? Suggest a reason for this difference.

Digestion

Digestion takes place when ingested food is broken down into a form that the body can use.

Digestion takes place in the mouth, stomach and small intestine. There are two types of digestion: mechanical or physical digestion and chemical digestion.

Mechanical digestion and its importance

The teeth are used to bite, tear and chew food into smaller pieces. The pieces need to be small enough to swallow. The food is chewed into smaller and smaller pieces. The breaking down of food into smaller pieces is called mechanical digestion.

In the stomach, the muscular wall contracts and relaxes to churn the food. The churning breaks the food into even smaller pieces. The function of mechanical digestion is to break the food into smaller pieces to provide a large **surface area**. This is very important for the next process, which is chemical digestion.

Word help

enzyme: a substance produced by a living organism which acts as a catalyst in biochemical reactions

digestion: the breakdown of larger insoluble molecules into smaller soluble ones

insoluble: cannot dissolve in water

soluble: can dissolve in water

surface area: the area of the outer part of something

The relationship between surface area and size

The concept of the relationship between the surface area and size is important in science. It helps us understand the rate at which certain processes take place.

The surface area is the total area of visible surfaces of an object and may be measured in cm^2 . Study Figure 3.3. The diagram shows how the breaking up of a solid into smaller pieces increases the surface area. An increased surface area of a substance increases the rate of a reaction.

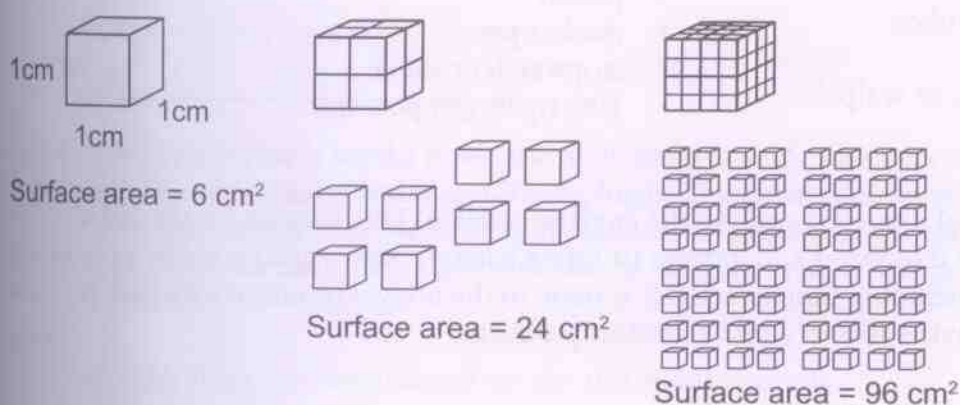


Figure 3.3 Breaking up a solid into smaller pieces increases the surface area

Chemical digestion and its importance

Mechanical digestion increases the surface area of the food for **enzyme** action during chemical digestion. Chemical digestion involves the action of digestive enzymes, which break the food down through chemical reactions. Chemical digestion starts in the mouth, continues in the stomach and is completed in the small intestine. Chemical digestion is important as it breaks the large, complex, **insoluble** molecules down into smaller, simpler, **soluble** molecules.

The soluble molecules can be absorbed into the blood and then into body cells. You will investigate chemical digestion in Activity 3.6.

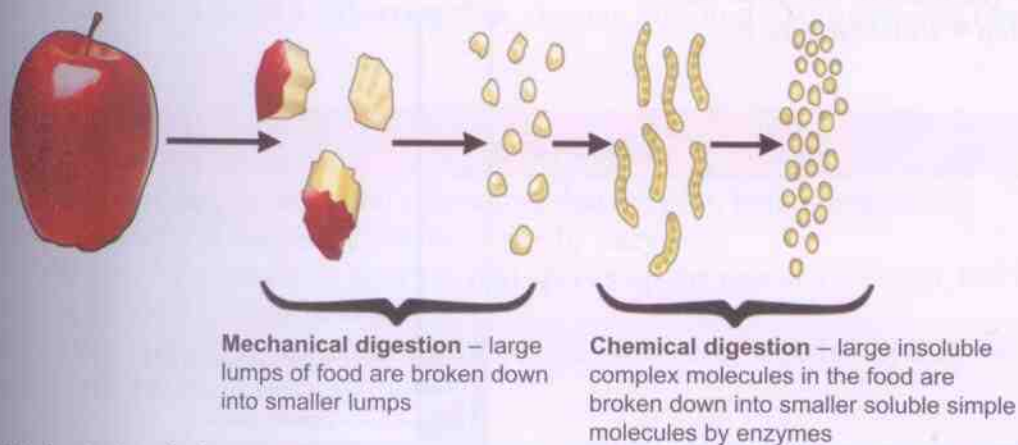


Figure 3.4 Mechanical and chemical digestion

Activity 3.3 Practical task

The effect of a larger surface area on the rate of a reaction

Aim: To investigate how an increase in surface area affects enzyme action

You will need:

- test-tube rack
- two large test tubes
- 10 ml syringe
- vegetable knife or scalpel
- white tile
- ruler
- potato
- marker pen
- stopwatch or timer
- 10% hydrogen peroxide

All plant and animal materials contain the enzyme catalase. This enzyme breaks down the chemical hydrogen peroxide into water and oxygen (it forms a foam). Hydrogen peroxide is produced in plant and animal cells, including human cells. It is toxic to the body. The enzyme catalase is produced to break down this toxic substance into harmless products.

Method

Follow the instructions.

1. Label the two test tubes A and B respectively.
2. Use the cutting tile and knife to cut two small potato cubes of about 7 mm x 7 mm x 7 mm. The cubes must be the same size.
3. Place one cube at the bottom of test tube A. Cut the other cube into as many small pieces as you can. Add the cut up cube to the bottom of test tube B.
4. Add 5 ml of 10% hydrogen peroxide to test tube A. Use the stopwatch to time how long it takes for the bubbles to stop forming. Record the time taken in the table. Measure the height of the foam that formed. Record the height of the foam in the table below.
5. Repeat step 4 with test tube B.

Safety box

Safety box: Hydrogen peroxide is a corrosive chemical. Do not let it touch your skin. Watch out for the bubbles when adding the hydrogen peroxide to the potato.

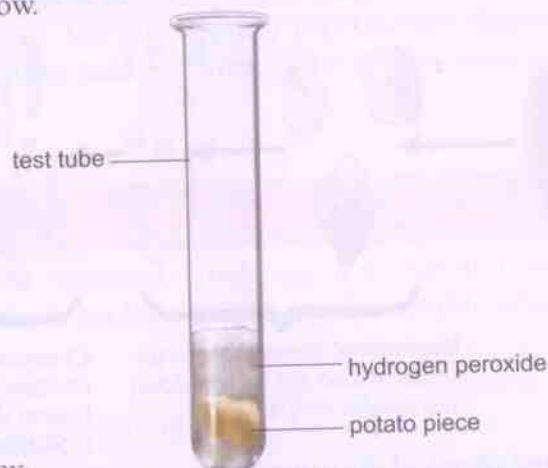


Figure 3.5 Potato pieces in hydrogen perox-

Observation and result

Record the results in a table.

Table 3.1: How an increase in surface area affects enzyme action

Potato	Reaction time (s)	Height of foam (mm)
Single cube		
Cut up cube		

The catalase in the potato pieces breaks down the hydrogen peroxide into water and oxygen. The gas released appears as many small bubbles, forming foam. The greater the surface area, the more the enzyme comes into contact with the hydrogen peroxide and therefore the quicker the reaction. The foam will also be higher in the test tube with the cut-up cubes due to the greater surface area.

Conclusion

Write a conclusion for this experiment based on the aim and the results.

Function of enzymes

An enzyme is a protein **molecule** manufactured in small quantities by a cell. It is a **catalyst** in many **biochemical** reactions that take place in the body. The chemical reactions that take place in living organisms are called metabolic reactions. Enzymes control metabolic reactions. Without enzymes the rate of metabolic reactions would be too slow.

Enzymes are catalysts that speed up (accelerate) the rate of the metabolic (biochemical) reaction, but do not undergo any change themselves during the reaction. When a chemical reaction is accelerated by a catalyst we call the reaction **catalysis**.

In a chemical reaction reactants change to form products. In a metabolic reaction where an enzyme acts as catalyst a reactant is called a **substrate**. The enzyme acts on the substrate and changes it into end products.

Word help

biochemical: the chemical processes and substances that occur in living organisms

catalysis: the breakdown of a substance or molecule by enzymes

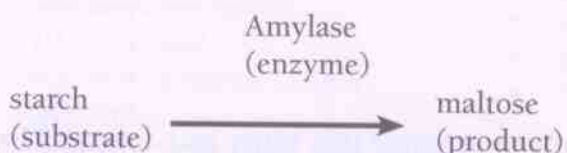
catalyst: a substance, for example an enzyme, that speeds up the rate of a reaction, but does not undergo any change itself during the reaction

molecule: a particle of matter consisting of two or more atoms held together by a chemical bond

substrate: the food that enzymes break down

During chemical digestion small insoluble pieces of food that are made up of large complex molecules are broken down into smaller, simpler molecules that are soluble. The enzymes in our alimentary canal are digestive enzymes. The digestive enzymes help break down larger molecules of protein, starch, and fat during digestion.

For example, in the digestive system the enzyme amylase is found in saliva in the mouth and pancreatic juice in the small intestine. Amylase acts on starch (a carbohydrate) as substrate and breaks it down into the product maltose (a sugar).



The maltose can then be broken down further by another enzyme to form glucose molecules.

Enzymes are specific in their action. This means that each kind of enzyme catalyses only one kind of reaction. For example, the enzyme amylase catalyses only the breakdown of starch into maltose (sugar). It will not break down any other kind of substrate.

The end products of digestion

The digestive enzymes break down specific larger molecules of protein, carbohydrates (starch and sugar) and fat (lipids) during chemical digestion.

The end products of digestion are summarised in the table.

Table 3.2 The end products of digestion

Food group	End product
carbohydrates	glucose
fats	fatty acids and glycerol
proteins	amino acids

The end products of digested large food molecules are soluble and small enough to be absorbed from the small intestine and then used in the body.

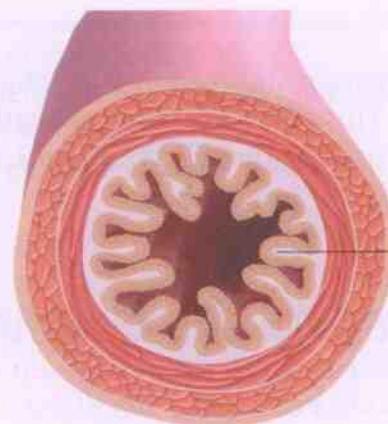
Absorption

Once food has been digested, the small soluble molecules are absorbed into the blood. **Absorption** occurs when soluble food molecules from the intestine are taken up or pass into the blood vessels. The process of absorption takes place through the walls of the small intestine.

Word help

absorption: a process by which molecules move through the wall of the small intestine into the blood vessels

The walls of the small intestine are folded and therefore have a large surface area for the absorption of nutrients. Figure 3.6 shows the increased surface area of the walls of the small intestine. The increased surface area allows for more molecules to be absorbed. The small intestine walls contain many blood vessels to transport the absorbed nutrients to all the cells in the body.



Folds in the wall increase the surface area for absorption

Figure 3.6 Cross-section of the small intestine to show the folds of the wall

Activity 3.4 Work in pairs

Study the diagram in Figure 3.7. Answer the questions.

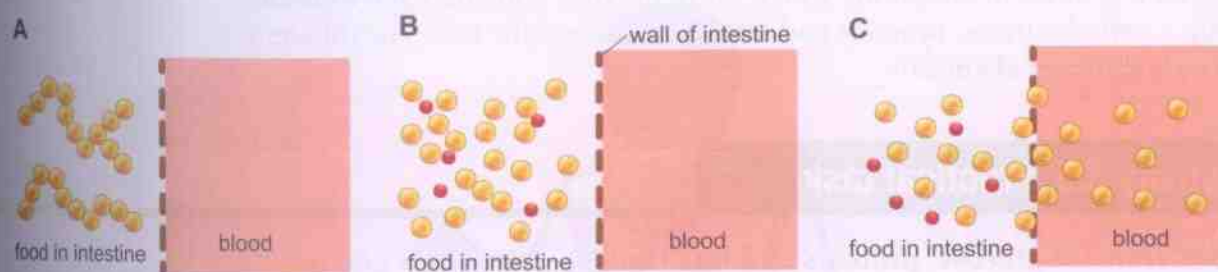


Figure 3.7 Why must food be digested?

- A food in intestine, blood
- B food in intestine, blood, wall of intestine
- C food in intestine, blood

1. Compare the food molecule in A with those in B. Describe how they are different.
2. What is present in B that has changed the food?
3. Describe what is happening to the food in C?
4. Explain what must happen to food molecules before they can be absorbed?

Assimilation

Once nutrients have been absorbed into the blood they are transported to all the cells where the nutrients are used. This process is called **assimilation**. Assimilation is the process whereby nutrients move into, and are used by the cells. In Form 1 you learnt about the functions of different nutrients in the body.

Egestion

Egestion is the process when undigested food is released from the body as faeces. Not all the food we eat is digested or absorbed into the body. Food that is not digested or absorbed remains in the alimentary canal and moves through the large intestine and the rectum. In the rectum faeces is formed. The faeces pass out of the body through the anus. This process is called **egestion**.

Word help

assimilation: when nutrients move into, and are used in, the cell

egestion: the elimination of undigested and unabsorbed food and nutrients from the digestive system

Food tests

You can test foods to determine which nutrients they contain. Each nutrient group – carbohydrates, proteins and lipids – has a specific test. The different tests use different chemicals.

Activity 3.5 Practical task

Food tests for glucose, proteins and fats (lipids) Work in a group

Part 1

Test for the presence of glucose

Aim: To test for the presence of glucose using Benedict's solution

You will need:

- test tube
- test-tube rack
- test-tube holder
- burner or beaker with boiling water
- syringe
- 2% glucose solution
- Benedict's solution

Method

Follow the instructions.

1. Use the syringe to add 2 ml of glucose solution to the test tube.

2. Rinse the syringe and add 2 ml of Benedict's solution to the test tube.
3. Heat the test tube over a Bunsen burner or in a water bath with boiling water for about a minute. Use the test-tube holder to move the hot test tube.
4. An orange-red colour will indicate the presence of glucose. This test can also be used to test for maltose.

Observation and results

Note the colour at the beginning of the experiment. Describe the colour change that you observed.

Conclusion

Write a conclusion based on the results.

Part 2

Test for the presence of protein

Aim: To test for the presence of proteins using the Biuret test

You will need:

- test-tube rack
- test tube
- syringe
- dropper
- 10% sodium hydroxide
- 1% copper sulfate solution
- 2 ml egg white

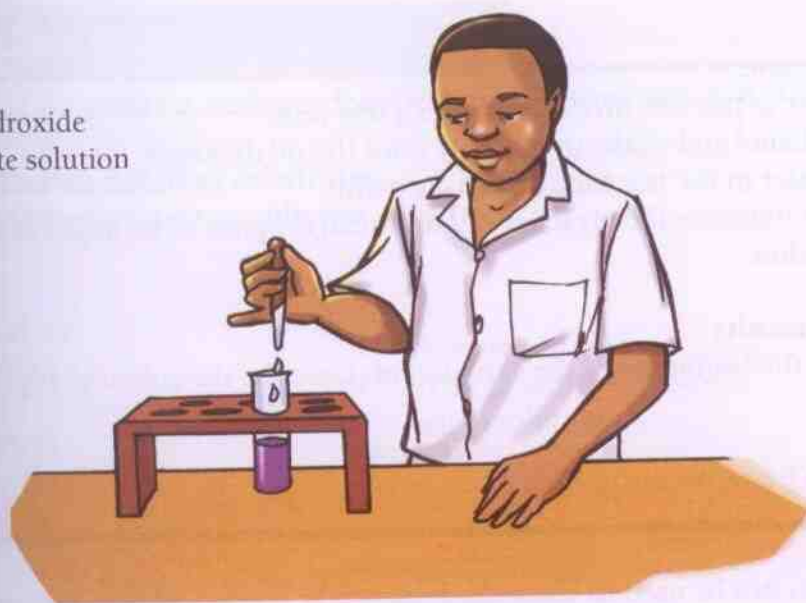


Figure 3.8 Testing for protein using the Biuret test.

Method

Follow the instructions.

1. Use the syringe to add 2 ml of egg white to a test tube.
2. Wash the syringe and add 2 ml of 10% sodium hydroxide solution to the egg white.
3. Use the dropper to add 3 drops of 1% copper sulfate solution.
4. Mix gently.
5. A lilac or purple colour will indicate the presence of proteins.

Observation and results

Note the colour at the beginning of the experiment. Describe the colour change that you observed.

Conclusion

Write a conclusion based on the results.

Part 3

Test for the presence of fats (lipids)

Aim: To test for the presence of fats (lipids) using ethanol

You will need:

- test-tube rack
- test tubes
- syringe
- dropper
- cooking oil
- water
- ethanol

Method

Follow the instructions.

1. Use the dropper to put two drops of oil into a test tube.
2. Add 5 ml of ethanol and shake thoroughly until the oil dissolves.
3. Add 5 ml of water to the test tube and shake gently.
4. A milky colour indicates the presence of fats. Small droplets of fat might also be visible. This is called an emulsion.

Observation and results

Note the colour at the beginning of the experiment. Describe the colour change that you observed.

Conclusion

Write a conclusion based on the results.

These three food tests can be used to show the presence or absence of carbohydrates (glucose), proteins and fats (lipids) in food samples. If the nutrient is present we say the result is positive.

Below is a table summarising the three food tests as well as the test for starch that you did in Topic 2

Table 3.3: Summary of main food tests

Test for?	Name of chemical	Is heating needed?	Description of a positive result
Glucose	Benedict's solution	Yes	Orange-red
Proteins	Biuret test: sodium hydroxide and copper sulfate	No	Lilac/purple
Fats (lipids)	ethanol	No	White emulsion
Starch	iodine	No	Blue-black

Action of enzymes on different food substrates

You can investigate the action of enzymes on different foods:

Protease enzymes like pepsin break down proteins into smaller molecules.

Lipase enzymes break down fats (lipids) into fatty acids and glycerol.

Amylase breaks down starch into maltose (simple sugar).

Activity 3.6 Practical task

Experimenting using enzymes on different food substrates Work as a class, Teacher demonstration

The action of amylase on different substrates

Aim: An experiment using amylase on different substrates

You will need:

- three 250 ml beakers
- three boiling tubes
- visking tubing
- three thermometers
- droppers
- amylase enzyme
- 2% starch solution
- 10% egg-white solution
- iodine solution

Method

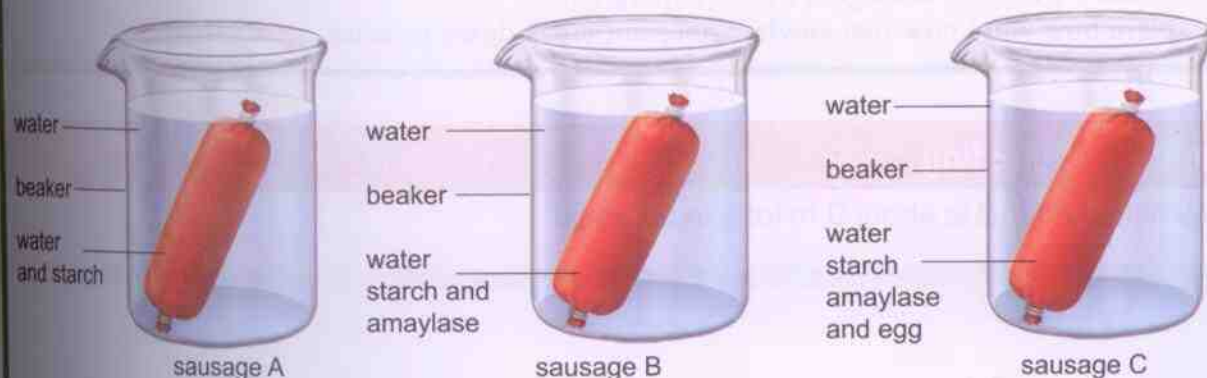


Figure 3.9 Experiment to show the working of amylase on different substrates

Follow the instructions.

1. Use the visking tubing to make three sausage-like bags as seen in Figure 3.11. Sausage A must contain starch solution, sausage B contains starch solution and amylase and sausage C must contain egg-white solution and amylase. The proportion of amylase to starch or egg-white solution should be about 5:1. This ratio means five parts of amylase to one part of starch or egg.
2. Use the thermometers to maintain the water bath between 30 °C and 35 °C.
3. Leave the containers for at least two hours.
4. After two or more hours test the water in beakers A and B for maltose using the Benedict's test. Test the water in beaker C for amino acids using the Biuret test.
5. If digestion takes place the small, soluble end products of digestion will be able to move out of the visking tubing and into the water. The water will then test positive for the presence of these end products. Large molecules will NOT be able to move through the tubing.

The expected results are in Table 3.4.

Table 3.4: Expected results of amylase test

Content of sausage	Result after food test	Digestion took place
Starch solution	Negative for maltose	no
Starch solution and amylase	Positive for maltose	yes
Egg white and amylase	Negative for proteins	no

Conclusion

Write a conclusion for the experiment based on the results.

Questions

Answer the questions.

1. Suggest why the sausages were kept at the same temperature.
2. a) Describe the effect of amylase on starch.
b) Give evidence of this effect from the experiment.
3. Why did we include sausage A in this experiment?
4. Explain how you know that amylase does not break down proteins.

Something interesting

The alimentary canal is about 9 m long in an adult!

Summary

- The human alimentary canal is a long tube for processing food; it includes and starts at the mouth (ingestion), the oesophagus, the stomach (digestion), the small intestine (digestion and absorption), the large intestine and ends in the anus (egestion).
- The gall bladder, pancreas and liver are associated with the alimentary canal.
- The pancreas makes enzymes for digestion in the small intestine.
- There are four types of teeth in a human. Each has its own function:
 - Incisors for cutting and biting food.
 - Canines for gripping and tearing food.
 - Premolars and molars for grinding and chewing food before swallowing.
- Mechanical digestion is the physical breakdown of food into smaller pieces.
- Chemical digestion is the breakdown of food through the action of enzymes.
- Amylase is an enzyme that breaks down insoluble starch into soluble maltose.
- All breaking-down enzyme reactions are called catalysis (breaking down).
- Digestion results in three kinds of end products:
 - Carbohydrates → glucose
 - Proteins → amino acids
 - Fats → fatty acids and glycerol

Topic assessment

Answer the questions.

1. a) Identify the labelled structures A to E.

[5]

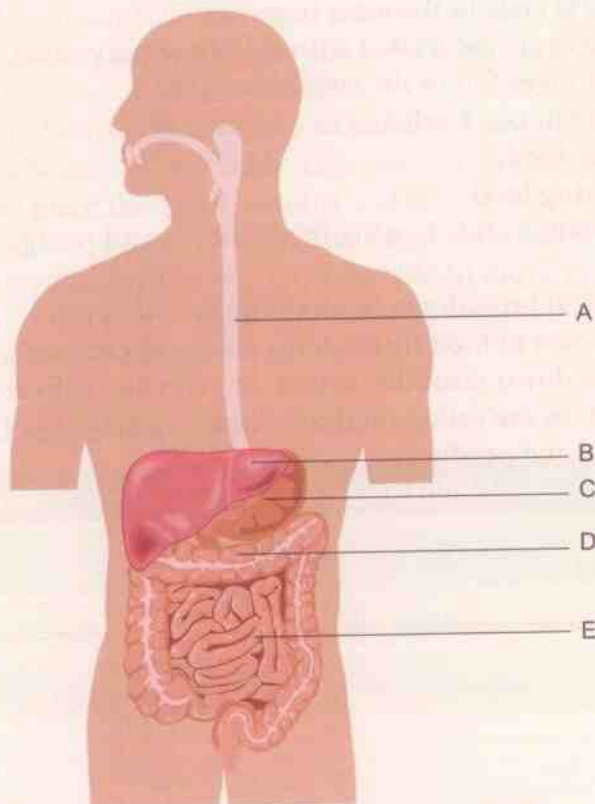


Figure 3.10 the human digestive system

[5]

- b) What are the functions of: (i) the mouth (ii) the anus [2]
c) Name the types of teeth used for grinding and chewing. [1]
d) Describe three ways in which you think the structure of the digestive system is adapted to its function. [3]
2. Explain the difference between mechanical digestion and chemical digestion. [2]
3. Name the end product from the digestion of:
a) protein [2]
b) lipids

Total marks [15]

Topic 4 Respiratory systems

Learning objectives and activities

Learning objectives	Activities
State the differences between inhaled and exhaled air	Carrying out an experiment to show the change in proportions of carbon dioxide and oxygen in inhaled and exhaled air
Describe the role of the alveoli in gas exchange	Describing gaseous exchange in the alveoli
Explain how the alveoli are adapted for gas exchange	Discussing adaptations

In Form 1 you learnt about the composition of the air and about respiratory gases. In Form 2 you learnt about the parts of the human respiratory system and the breathing mechanism (ventilation).

Inhaled and exhaled air

The atmospheric air you **inhale** consists of 21% oxygen, 78% nitrogen and about 0.03% carbon dioxide. When air enters the lungs, some of the oxygen is removed from it and absorbed into the body. Some carbon dioxide is added to the air to be exhaled. The air that is exhaled will therefore have more carbon dioxide and less oxygen. The air that you **exhale** will also have slightly more water vapour. The air is moistened when it moves through your lungs and nasal passages on its way out. Oxygen and carbon dioxide are respiratory gases because they are involved in respiration. You learnt about respiration in Topic 2 Plant nutrition.

Table 4.1 The composition of inhaled and exhaled air.

	Inhaled air %	Exhaled air %
nitrogen	78	78
oxygen	21	16
carbon dioxide	0.03	4
water vapour	less	more

In Form 1 you tested for the presence of carbon dioxide by blowing into lime water. You learnt that lime water turns milky in the presence of carbon dioxide.

Word help

exhale: breathe out

inhale: breathe in

Activity 4.1 Practical task

The difference in composition between inhaled and exhaled air

Work in a group

Aim: To compare the composition of inhaled and exhaled air

You will need:

- two conical flasks
- glass tubing
- rubber tubing
- rubber stoppers
- clear lime water

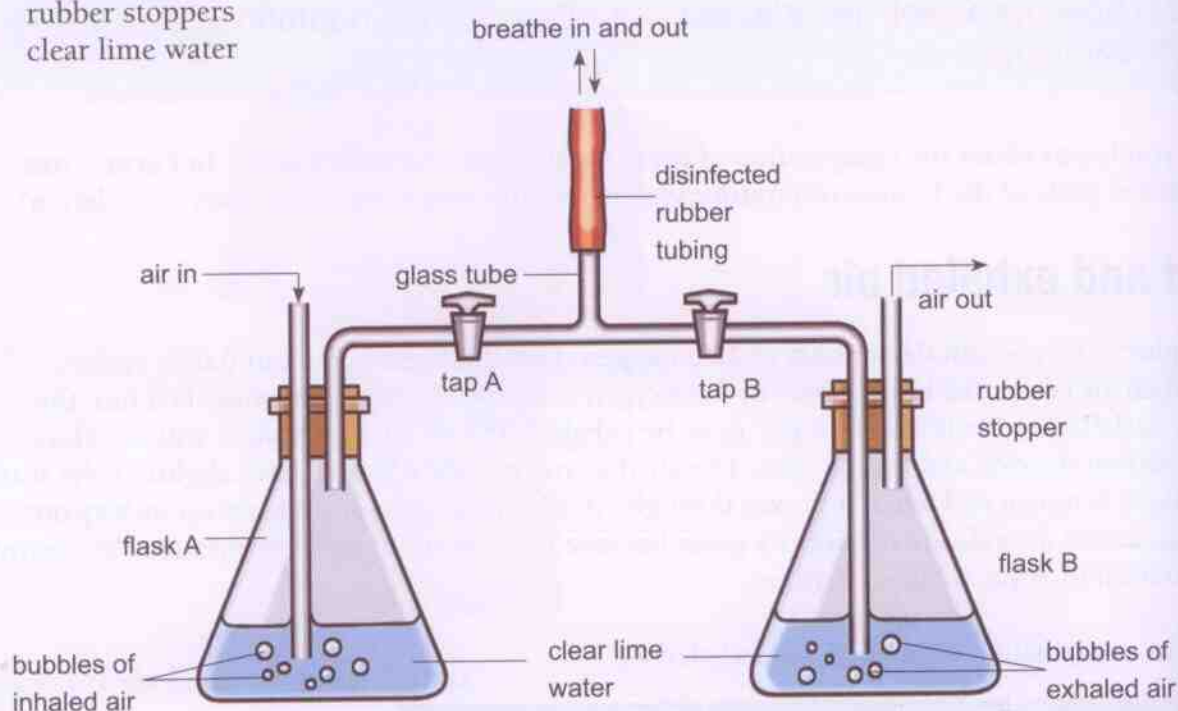


Figure 4.1 An experiment showing the difference between inhaled and exhaled air

Method

Follow the instructions.

1. Set up the apparatus as indicated in Figure 4.1. Make sure that the glass tubes are in the correct position.
2. Breathe in and out through the disinfected rubber tubing. As you inhale, air from the atmosphere will bubble through flask A. As you exhale, your exhaled air will bubble through flask B.
3. Observe the colour changes as air bubbles through the clear lime water.
4. Air containing higher levels of carbon dioxide will turn the lime water from a clear to a milky colour.

Use the Internet to find a demonstration of this experiment.

Observations and results

Copy and complete the table with the results.

Table 4.2: The colour of lime water in flasks A and B

Flask	Colour of lime water
A	
B	

Conclusion

Write a conclusion based on the results.

Questions

Answer the questions.

1. Explain why the lime water in flask A remained the same and the lime water in flask B changed colour.
2. From this experiment what can you conclude about the composition of inhaled and exhaled air?

In Form 1 you carried out an investigation to compare the percentage of oxygen in inhaled and exhaled air. It is not so easy to determine the percentage of carbon dioxide in air as it is less than 1%.

The human respiratory system

In Form 2 you learnt the parts of the respiratory system. Air enters your body through your nose or mouth. It then moves through the trachea and enters your lungs. In the lungs the air moves through tubes called bronchi (sing. bronchus) and then smaller tubes called bronchioles (sing. bronchiole). The air then reaches tiny air sacs called alveoli (sing. alveolus). The alveoli are surrounded by a network of capillaries. Figure 4.2 shows the structure of the human respiratory system.

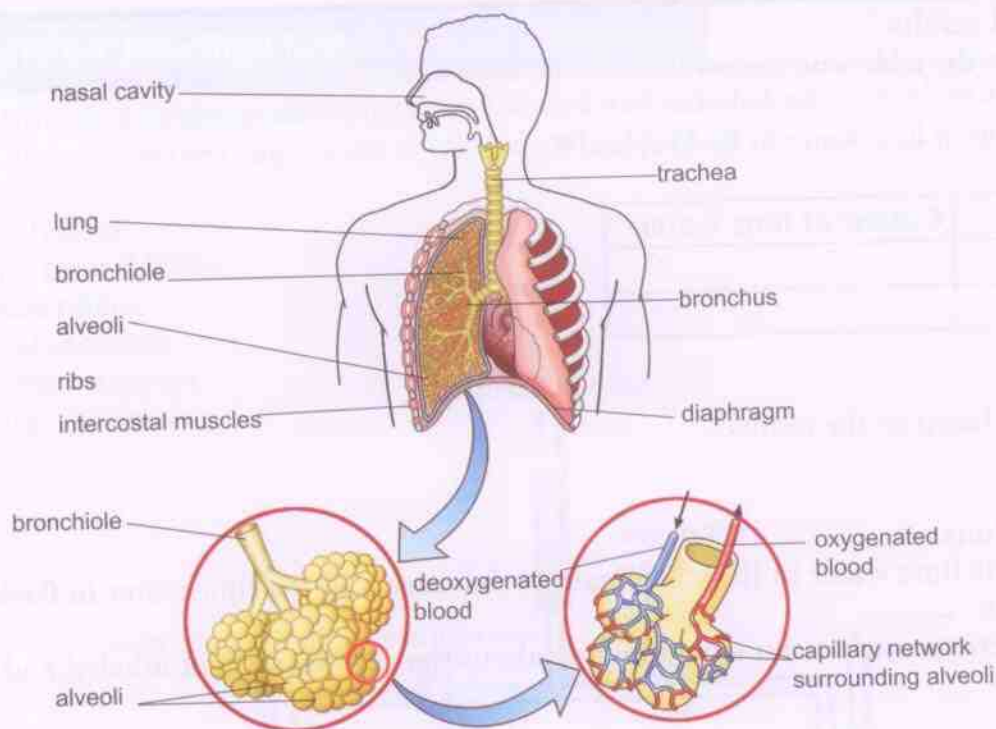


Figure 4.2 The human respiratory system

The structure of the alveoli

The structure of the alveoli is adapted to the function of **gaseous exchange**. The bronchioles end in microscopic air sacs, the alveoli. Each lung has millions of microscopic air sacs, the alveoli. All the alveoli together create a very large surface area for the exchange of the respiratory gases. The large surface area allows for oxygen to move into the blood and carbon dioxide to be removed from the blood.

The walls of the alveoli consist of one cell layer so that gases can diffuse easily through the wall of the alveoli into and from the blood. The inside of the alveoli is moist to keep the alveoli open and so that oxygen can dissolve. This allows for easier diffusion. Around each alveolus is a network of blood capillaries. This allows for effective transport of the respiratory gases to and from the lungs. See Figure 4.3.

Gaseous exchange in the alveoli

The exchange of gases in the alveoli in the lungs is called gaseous exchange.

Word help

gaseous exchange: the exchange of gases through a membrane.

The air entering the alveoli has a higher concentration of oxygen than the blood in the capillaries. Oxygen diffuses from the alveoli into the blood from a high concentration to a low concentration. The oxygen is then transported to all the cells in the body to be used for respiration.

The carbon dioxide produced by respiration is transported to the lungs. The blood surrounding the alveoli has a higher concentration of carbon dioxide than the inhaled air in the alveoli. The carbon dioxide diffuses from the blood and into the alveoli to be exhaled.

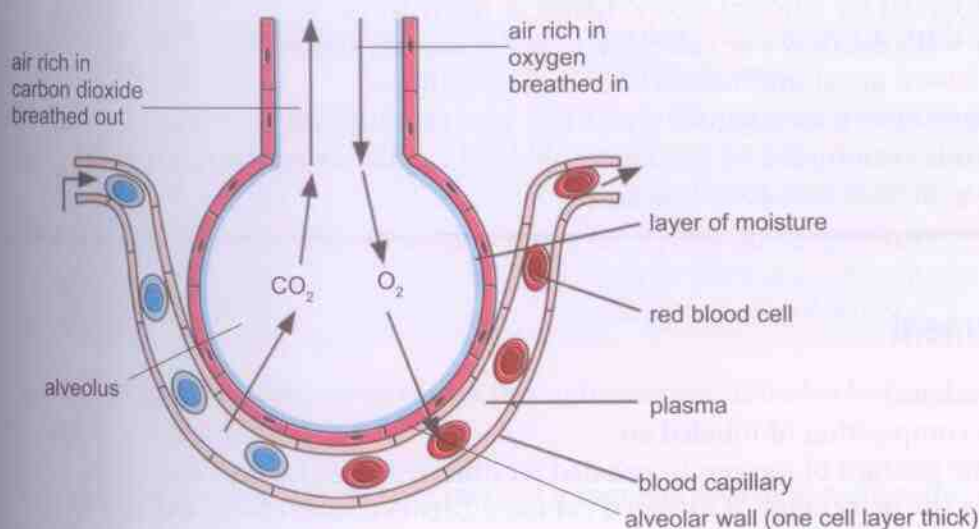


Figure 4.3 Gaseous exchange in an alveolus

Activity 4.2 Work in pairs

Answer the questions

1. Use Figure 4.3 to answer the questions.
 - a) Suggest how the number of alveoli can help the lungs get more oxygen into the blood.
 - b) Suggest why there are many capillaries around the alveoli.
 - c) Use the diagram to describe gaseous exchange in your own words.
2. List the parts that carbon dioxide moves through, from the blood to the atmosphere.

Smoking breaks down the walls of the alveoli. This means less surface area for the diffusion of oxygen and carbon dioxide.

People with this problem have emphysema. Breathing for these people is VERY difficult as they do not get enough oxygen from the air. You will learn more about how smoking damages the alveoli in Topic 9 Health and diseases.

Summary

- Inhaled air has 78% nitrogen, 21% oxygen and 0.03% carbon dioxide.
- Exhaled air has 78% nitrogen; 16% oxygen and 4% carbon dioxide.
- Exhaled air contains more water vapour than inhaled air.
- Diffusion of oxygen and carbon dioxide takes place in the alveoli – microscopic air sacs deep inside the lungs.
- Alveoli are adapted for efficient gas exchange in many ways.
- The alveolar walls are only one cell thick to allow for easy diffusion.
- The alveoli have a moist internal surface to allow diffusion.
- There are many alveoli forming a large surface area for diffusion.
- Each alveolus is surrounded by a network of blood capillaries for the transport of gases. They release carbon dioxide and absorb oxygen.

Topic assessment

Answer the questions.

1. a) State the composition of inhaled air. [3]
b) How is the amount of oxygen in exhaled air different from inhaled air? [1]
c) How does the percentage of nitrogen compare between inhaled and exhaled air? [2]
Suggest a reason for this. [1]
2. a) Which chemical is used to show the presence of carbon dioxide in air? [1]
b) Describe how the chemical named in 2.a) changes if carbon dioxide is added. [1]
3. a) Where does gaseous exchange take place? [1]
b) Explain how TWO adaptations of the alveoli help gaseous exchange. [2]

Total marks [11]

Topic 5 Transport in plants

Learning objectives and activities

Learning objectives	Activities
• Explain the process of transpiration • State the factors affecting the rate of transpiration	• Discussing the process of transpiration • Carrying out experiments to investigate factors affecting the rate of transpiration
• Measure transpiration in a plant • Outline the importance of transpiration	• Explaining the importance of transpiration
• Explain the terms plasmolysis and turgidity	• Discussing plasmolysis and turgidity • Watching video simulations on plasmolysis and turgidity
• Describe the effects of water loss and water gain in plant cells	• Carrying out experiments to demonstrate plasmolysis and turgidity using potato strips

In Form 1 you learnt that water moves in plants through osmosis. You also learnt about the process of diffusion by which particles move.

In Form 2 you learnt about the internal structures of a root and a stem as well as the water and ion uptake by plants.

In Topic 2 Plant nutrition, you learnt about the internal structure of the leaf.

Transport in plants

Plants are multicellular and need a system to transport nutrients and water to all cells. In Form 2 you learnt that in plants, xylem transport water and ions (minerals). Phloem, however, transports glucose produced by photosynthesis. In Form 1 you learnt that water from the soil enters a plant through its root hair cells by osmosis. In Form 2 you learnt that ions from the soil enter the plant through active uptake. Once inside the root the ions are dissolved in water and move with the water through the plant. The process that helps water to move through a plant is called **transpiration**.

Transpiration

Transpiration is the loss of water vapour from the surface of the leaves by evaporation, which results in more water being drawn up from the roots. The water vapour leaves the leaf through the stomata (sing. stoma). In Topic 2 Plant nutrition you learnt that stomata are openings on the bottom surface of the leaf. Stomata are surrounded by guard cells that can open and close the stomata.

Word help

transpiration: the loss of water vapour from the leaves of a plant

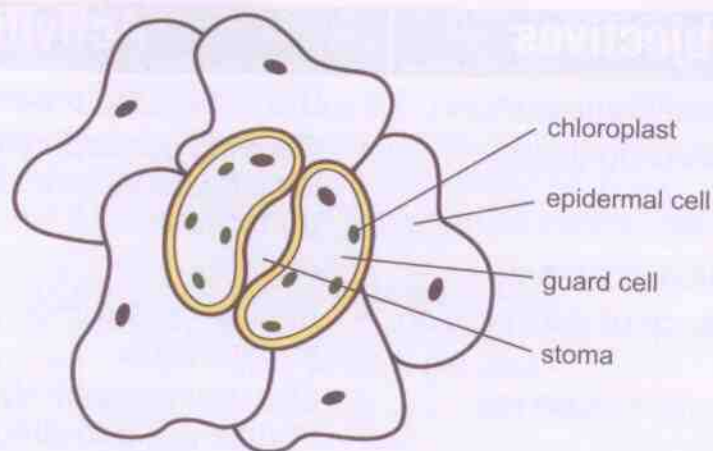


Figure 5.1 A stoma in the lower epidermis of a leaf

When water from the stem, enters a leaf in the xylem of the veins, it moves from the xylem to the cells of the leaf through osmosis. When the water reaches the air spaces between the spongy mesophyll in the leaf close to the stomata, it evaporates from the cells into the air space as water vapour.

This water vapour then diffuses from the air space out of the leaf through the stomata. The water vapour diffuses from a high concentration of water molecules inside the leaf of the plant to a lower concentration of water molecules outside the leaf in the air. As the water moves out of the leaf it has a pulling effect on the rest of the water in the leaf and xylem of the plant. This effect is known as transpiration pull and plays a role in the movement of water through a plant from the roots to the leaves.

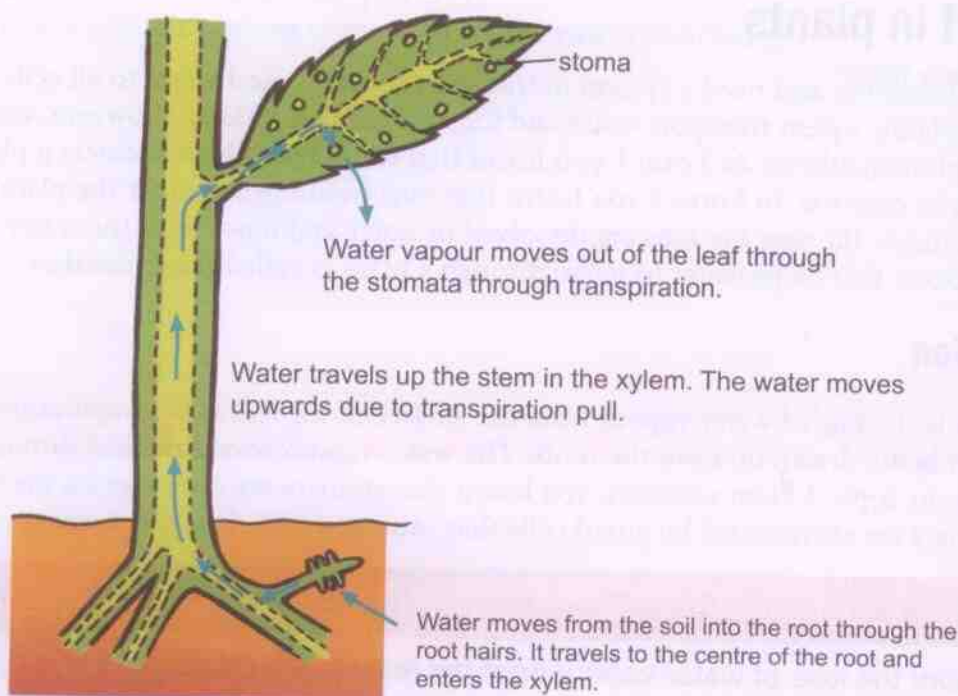


Figure 5.2 Movement of water through a plant

Transpiration pull is not solely responsible for the movement of water upwards in a plant, **root pressure** also assists with moving water in the plant. As water enters the roots through osmosis the xylem vessels fill up and this puts pressure on the more rigid outer cells of the root. This pressure causes the water to be forced up the plant, despite the force of gravity.

Factors affecting the rate of transpiration

Any factor that influences the **rate** of evaporation and water vapour movement, will also affect the rate of transpiration in plants.

Environmental conditions like temperature, **humidity** and wind have an influence on the rate at which transpiration takes place. Higher temperatures and wind increase the rate of transpiration as both these factors increase the movement of water molecules away from the leaf. This causes the air surrounding the leaf to have a low concentration of water molecules and the rate of transpiration is increased. The higher the humidity in the air, the lower the rate of transpiration. At a high humidity the concentration of water particles in the air surrounding the leaf is high. Therefore the rate of transpiration is lower. You will investigate these factors later in the topic using a potometer. A **potometer** measures the amount of water uptake. The amount of water taken up by the plant is dependent on the transpiration rate. The faster the transpiration rate the more water will enter the plant.

Other factors that affect the rate of transpiration are the size of the surface area from which transpiration is taking place and the number of stomata on a leaf. The larger the surface area from which transpiration can take place the faster it will be. The more stomata a leaf has, the higher the rate of transpiration. Light intensity plays a role in the opening and closing of stomata by affecting the guard cells. Because no transpiration can take place if the stomata are closed, light intensity also affects the rate of transpiration.

Activity 5.1 Practical task

Factors affecting the rate of transpiration. Work as a group

Aim: To investigate the effect of temperature, humidity and wind speed on the rate of transpiration using a potometer

You will need:

- a graduated potometer

Word help

rate: a certain quantity of one thing considered in relation to (per) another thing. For example: millimetres per second.

humidity: the amount of water vapour in the air

potometer: an apparatus used for measuring the rate of water uptake of a leafy plant shoot

root pressure: the pressure caused by osmosis of water into the roots and results in water (sap) to move through the plant to the leaves

- cutting tool
- petroleum jelly
- a stopwatch or timer
- transparent plastic bag big enough to cover the leafy shoot
- elastic band or adhesive tape
- a fan to create wind

- a large sink or basin of water
- a fresh, leafy shoot

Safety box

Take care when cutting the plant shoot. Take care assembling and handling the potometer: the long glass tubes break easily.

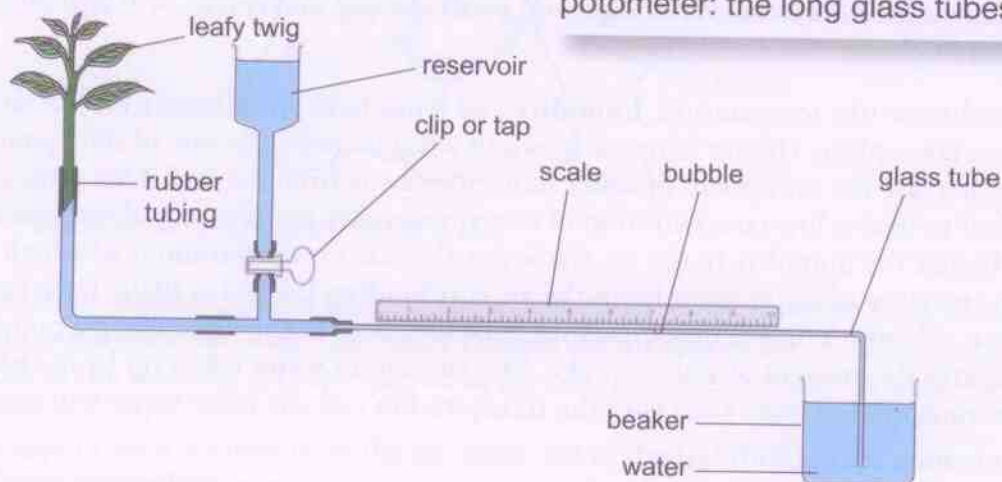


Figure 5.3 A potometer

Method

Follow the instructions.

1. Set up the leafy shoot in the potometer as shown in Figure 5.3. Precautions to take when setting up the potometer:
 - Make sure that the stem of the plant fits very tightly into the rubber tubing. Seal with petroleum jelly.
 - The leafy shoot should be cut under water and inserted under water to avoid any air bubbles. Air bubbles will block the xylem, preventing further water uptake.
 - The leaves must be dry.
2. Lift the open lower end of the graduated tube out of the water briefly to allow air to enter. Replace the tube in water when an air bubble can be seen in the tube.
3. When the bubble reaches the graduated part of the tube, measure the distance the bubble moves along the tube in one minute. As soon as you have taken the measurement, open the tap carefully to move the bubble back to the beginning of the tube. Do not allow the bubble to move into the plant.
4. Measure the distance twice and get an average. Record all the distances in a table. See Table 5.1.
5. Place the potometer in a warm sunny spot for about five minutes. Five minutes should be enough time for the plant to adjust to the new condition.
6. Repeat steps 3 and 4.
7. Place the potometer in front of a fan. Leave for five minutes.
8. Repeat steps 3 and 4.

9. Place a large plastic bag over the plant and secure with tape. Allow to stand for two hours or overnight. This will ensure that the leaves inside the bag will be in very humid conditions. You will notice, that after standing for a while, that water is visible on the inside surface of the plastic bag. This is proof that transpiration took place.
10. Repeat steps 3 and 4.

Observations and results

Table 5.1 Distance an air bubble moved along a potometer glass tube per minute in different conditions

Condition	Distance (mm)		
	1st	2nd	Average
Normal			
In the sun			
In wind			
In high humidity			

Conclusion

Write a conclusion based on the results.

Questions

Answer the questions.

1. Suggest how the results would differ if you used a desert plant for this experiment?
2. When you are investigating the effect of wind on the rate of transpiration, why is it important to keep the plant warm and in bright light at the same time?
3. Look at the results a learner got when investigating transpiration using a potometer.

Table 5.2

Movement of bubble (mm per minute)	
Still air	Moving air
12	24

Use the results to describe the effect that moving air (wind) has on the rate of transpiration.

The importance of transpiration

Transpiration is essential for a plant as it causes transpiration pull. This process pulls the water up from the roots, through the stem, into the leaves and out into the air. Look again at Figure 5.2. Follow the arrows which show the path the water moves through the plant.

Water enters the root hair cells by osmosis. Minerals in the form of ions also enter the root hair cells from the soil through active uptake. The water carries the minerals to all parts of the plant. The

minerals are needed to keep the plant healthy and to grow. Water is necessary for photosynthesis, to dissolve minerals and to keep the plant turgid. You will learn more about turgidity later in the topic. As water vapour exits the leaves, it cools the leaves by evaporation. It is like when you sweat and the sweat evaporates from your body causing you to cool down. The evaporation of the sweat makes your body cooler.

Plasmolysis and turgidity

Inside a plant water moves from cell to cell by osmosis. This is because there is a higher concentration of water molecules than the next, so the water moves through the plant to all the cells.

When a plant has enough water available, the cells and the large vacuoles inside the cells are filled with water. The vacuole pushes the cytoplasm against the cell wall and the cell is firm. The cell walls are strong and they do not break due to the water pressure. The cells will be firm. We describe such cells as being **turgid**. Turgid cells provide support to a plant. When all the cells in a plant are turgid the plant appears healthy and upright.

However when a plant lacks sufficient water, the cells and the large vacuoles will lose water by osmosis. As the vacuole loses water it gets smaller and does not push the cytoplasm against the cell wall anymore. The pressure inside the cell decreases. The cell membrane shrinks away from the cell wall. These cells are **plasmolysed**. The process that the cells undergo when losing water is called **plasmolysis**. The cell becomes limp. When the cells in a plant become plasmolysed the plant appears unhealthy and droopy. We say that the plant is **flaccid** (wilted).

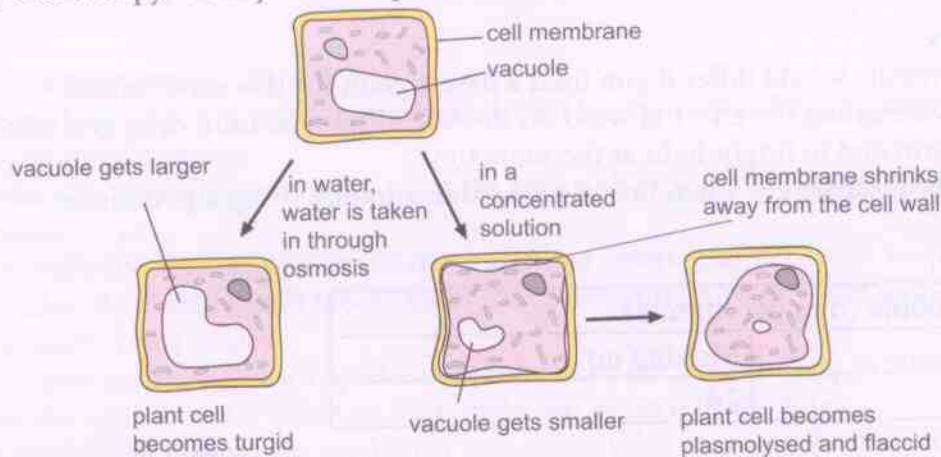


Figure 5.4 Turgid and plasmolysed cell

Word help

plasmolysis: the shrinking of the cell contents away from the cell wall of a plant cell due to water loss
flaccid: the condition of a cell when it has lost water and is limp and floppy

plasmolysed: a condition in which the contents of a plant cell have shrunk because of a loss of water by osmosis

turgid: the condition of a cell that it is firm and filled with water

turgidity: when a cell is fully expanded due to the absorption of water up to the limit; a cell is turgid when it is full of water and cannot take in any more

Look at the two plants in Figure 5.5. The plant in Figure 5.5a) has been given as much water as it needs. Its cells are turgid and hold the plant stems and leaves firm and up. The same plant in Figure 5.5b) has suffered severe water loss. Its cells have become plasmolysed. Unless such a plant is given a large amount of water it will die.

a) Turgid plant



b) Wilted plant



Figure 5.5 A turgid and wilted plant

Activity 5.2 Practical task

Demonstrating turgidity and plasmolysis using potato strips

Work with a partner

Aim: To investigate turgidity and plasmolysis in potato strips

- You will need:
- three 100 ml beakers
- cutting tool
- cutting tile
- timer
- electronic scale

- ruler
- 20% salt solution
- pure water
- potato
- paper towel

a) Potato strips in pure water



b) Potato strips in 20% salt water

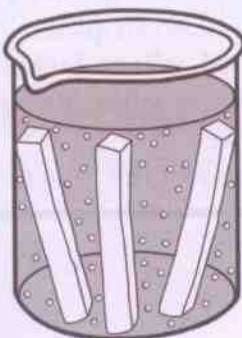


Figure 5.6 Investigation into turgidity and plasmolysis

Method

Follow the instructions.

1. Prepare two beakers for the potatoes. One should have 90 ml of distilled water and the other should have 90 ml of 20% salt solution.
2. Cut two potato strips of 5 mm wide, 5 mm high and 30 mm in length. Make sure that the potato strips are exactly the same size. Dry the potato strips.
3. Determine the mass of each strip and record the mass in a table. See Table 5.3.
4. Place one potato strip in each of the beakers and leave them for 20 min as in Figure 5.6.
5. Remove the strips from the beaker and dry them with paper towel. Be careful not to mix them up.
6. Measure the length of each potato strip accurately and record the length in the table.
7. Measure the mass of each strip accurately and record the mass in the table.
8. Feel the texture and degree of hardness or softness of each strip. Record this in the table.

Observations and results

Table 5.3 The length, mass and feel of two potato strips before and after being placed in water or salt solution

Type of liquid	Length (mm)		Mass (g)		Feel after
	before	after	before	after	
water					
20% salt solution					

Conclusion

Write a conclusion based on your results. Also mention which cells are turgid and which cells are plasmolysed, based on your results.

Questions

Answer the questions.

1. In which liquid did the potato strips lose mass and length? Explain why this happened.
2. In which liquid did the potato pieces gain mass and length? Explain why this happened.
3. Explain how the difference in how the potato strips felt at the end of the experiment helped you to determine which strip had plasmolysed cells and which strip had turgid cells.
4. Suggest what might happen to potato pieces left in air for the same length of time as your experiment. Give a reason for your answer.
5. What do you think might happen to many of the cells in a plant on a hot, dry day?

Summary

- Transpiration is the loss of water from the leaves of a plant.
- Various factors affect the rate of transpiration.
- Wind speed – the greater the wind speed the higher the rate of transpiration.
- Temperature – the higher the temperature, the higher the rate of transpiration.
- Leaf surface area – the greater the surface area the higher the rate of transpiration.
- Light intensity – the higher the light intensity the higher the rate of transpiration.
- The number of stomata – the more stomata the higher the rate of transpiration.
- Humidity – the lower the humidity the higher the rate of transpiration.
- Transpiration is important because:
 - it causes a stream of water to enter and leave the plant – allowing water and dissolved minerals to enter the root hair cells and reach all parts of the plant
 - the evaporation of water from the leaf mesophyll cells helps to cool the plant
- Plasmolysis is the shrinking of the cell contents away from the cell wall of a plant cell due to water loss. The plant becomes wilted when the cells lose water.
- Turgidity is when a cell is full of water. The plant is turgid when the cells are full of water.

Topic assessment

Answer the questions.

1. a) Define transpiration. [1]
b) Describe the importance of transpiration in a plant. [2]
2. a) Name two environmental factors that increase the rate of transpiration. [2]
b) Explain how these factors increase the rate of transpiration. [4]
3. a) State two precautions when using a potometer. [2]
b) A potometer really measures the water going into a plant. Is it fair to use it to measure transpiration? Why? [2]
4. a) Define:
(i) Plasmolysis [2]
(ii) Turgidity [2]
b) Describe the appearance of a plant if its cells become plasmolysed. [2]
c) Suggest what can be done to such a plant to get its cells turgid again. [1]

Total marks [20]

Topic 6 Transport system in humans

Learning objectives	Activities
Describe the double circulatory system	- Watching video simulations on the circulatory system - Examining a model of the human circulatory system

In Form 1 you learnt about the components of blood. In Form 2 you learnt about the heart structure and the main blood vessels to and from the heart.

Blood circulation

Blood circulates through your body in the circulatory system. The blood carries nutrients and oxygen to the cells and waste products from respiration and carbon dioxide away from the cells. The heart pumps the blood through your body to all the organs and cells.

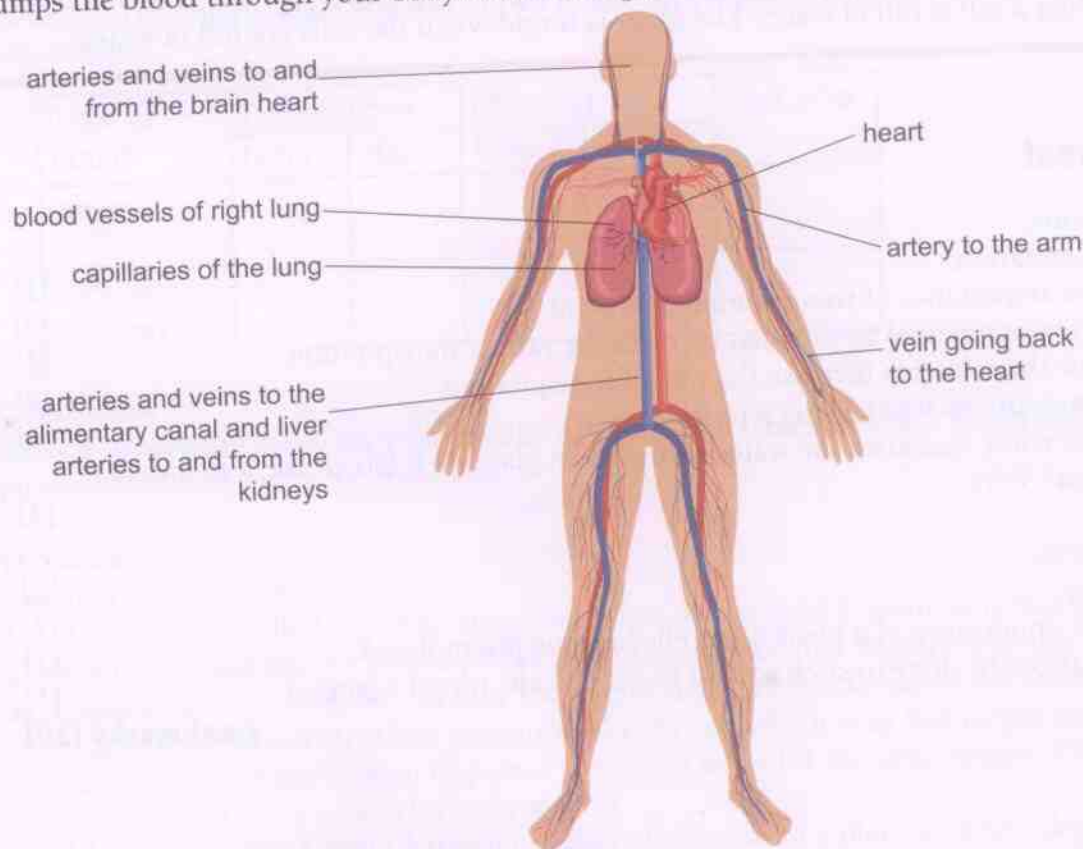


Figure 6.1 the human circulatory system

Word help

circulation: repeated movement in the same circuit or path.

double circulatory system: blood flows through the heart twice; consists of two circulatory systems

The double circulatory system

The human circulatory system has many blood vessels. The arteries carry blood away from the heart. The veins carry blood back to the heart.

Humans have a **closed circulatory system**, since blood is contained within blood vessels. For every complete journey around the human body, blood flows through the heart twice. The human circulatory system is therefore called a **double circulatory system**. The circulatory system consists of a **pulmonary circulatory system** and a **systemic circulatory system**.

Look at Figure 6.2. The pulmonary circulatory system transports blood from the heart to the lungs and back to the heart. The systemic circulatory system transports blood from the heart to all the body organs and cells and back to the heart.

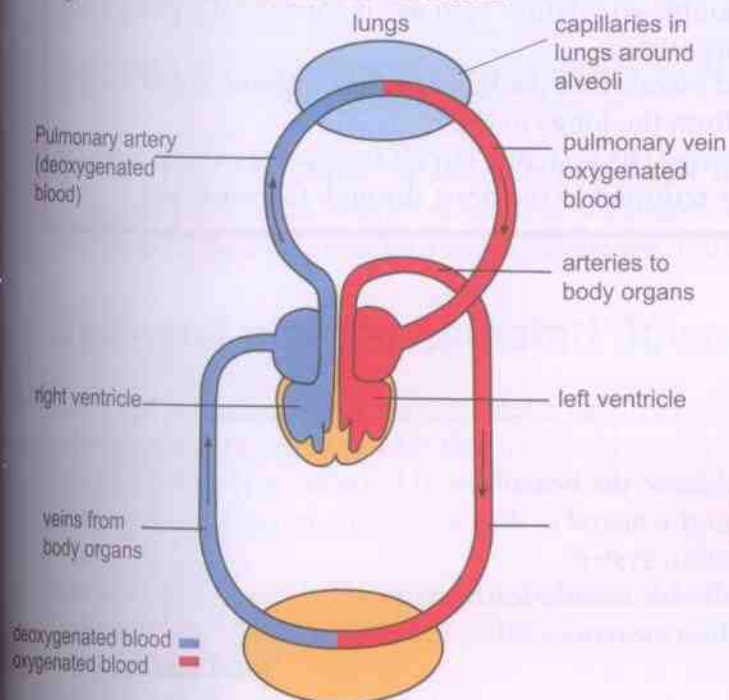


Figure 6.2 The double circulatory system

The pulmonary artery carries blood from the heart to the lungs. The blood in the pulmonary artery has very low concentrations of oxygen and is therefore **deoxygenated**. At the lungs the blood absorbs oxygen from the lungs and releases carbon dioxide. You learnt about gaseous exchange in Topic 4 Respiratory systems. The blood with a higher oxygen concentration is **oxygenated**. The pulmonary vein carries oxygenated blood from the lungs to the heart. This is the pulmonary circuit.

Something interesting

If all the blood vessels in the body were put end to end in a line it would be 100 000 km long!

It takes about 20 seconds for a red blood cell to circle the whole body. They move fast!

Word help

closed circulatory system: blood is contained within blood vessels

deoxygenated blood: blood containing low oxygen levels and high carbon dioxide levels.

oxygenated blood: blood containing high oxygen levels and low carbon dioxide levels.

pulmonary: regarding the lungs

pulmonary circulatory system: the system in which blood flows from the heart to the lungs and back to the heart

systemic circulatory system: the system in which blood flows from the heart to the body tissues and back to the heart

The oxygenated blood is then pumped through the body in the systemic circuit. Blood leaves the heart in the aorta. The systemic circuit transports oxygen to all the cells for respiration and removes carbon dioxide. Deoxygenated blood is returned to the heart through the veins. The veins join up to form the vena cava that enters the heart.

The blood is pumped through two circuits so that the flow is fast and under high pressure. It means that all the cells in all parts of the body receive a blood supply rich in oxygen and that all wastes can be removed. The two circuits happen at the same time. Our circulation system is very efficient.

Summary

- Humans have a closed circulatory system.
- The human circulatory system is called a double circulatory system - it consists of a pulmonary circulatory system and a systemic circulatory system.
- The pulmonary artery carries deoxygenated blood from the heart to the lungs but the pulmonary vein carries oxygenated blood from the lungs to the heart.
- Oxygenated blood leaves the heart in the aorta. The systemic circuit transports oxygen to all the cells for respiration. Deoxygenated blood is returned to the heart through the vena cava.

Topic assessment

Answer the questions

1. a) What is the function of the heart? [1]
b) Through which blood vessel does blood leave the heart? [2]
c) Which blood vessels carry blood back to the heart? [2]
2. a) Explain the meaning of a double circulation system. [2]
b) Name the two circulation systems in a double circulation system. [2]
c) State one advantage of a double circulation system. [1]

Total marks [10]

Topic 7 Reproductive systems in plants

Learning objectives and activities

Learning objectives	Activities
• Explain the structures of wind- and insect-pollinated flowers	• Examining the structure of wind- and insect-pollinated flowers • Drawing the structure of wind- and insect-pollinated flowers
• Describe the process of germination	• Discussing the process of germination
• Investigate conditions necessary for germination	• Carrying out experiments to demonstrate conditions necessary for germination
• Calculate percentage germination	• Determining germination percentage

In Form 1 you learnt about the structure of a simple flower, pollination and fertilisation. In Form 2 you learnt to distinguish between monocotyledonous (maize) and dicotyledonous (beans) plant seeds. You also described the functions of the cotyledon and the endosperm.

Wind- and insect-pollinated flowers

Plants and animals all have ways to make sure that they produce **offspring**. Without efficient pollination plants would not be able to reproduce. In Form 1 you learnt that pollination is the process by which pollen is transferred from the stamen (male part) to the carpel (female part) of a flower. Pollen grains contain the male sex cells. The female sex cell is located in the ovule of the ovary.

Flowering plants use different agents of pollination, for example, mammals, birds, the wind or insects for pollination. They can also self-pollinate.

Different flowers have adapted their structures to be suited to a specific pollinating agent.

Wind-pollinated flowers

Wind-pollinated flowers have adapted to be pollinated by the wind. Grasses are wind pollinated. You may not even know that grasses have flowers because of the lack of coloured petals.

Word help

offspring: the product or children produced by reproduction

Wind-pollinated flowers have to produce a lot of very light and dry pollen that can be carried long distances by the wind. The anthers and stigma are long so that they are exposed to the wind and not protected inside the petals of the flower. The petals are small. The pollen can then be caught in the wind and carried away. The stigmas of these flowers are often feathery to catch as much pollen from the wind as possible. Wind-pollinated flowers do not have to attract insects so their petals are not brightly coloured; the petals are green. The flowers also do not produce nectar or have a scent. Figure 7.1 shows a wind-pollinated flower.

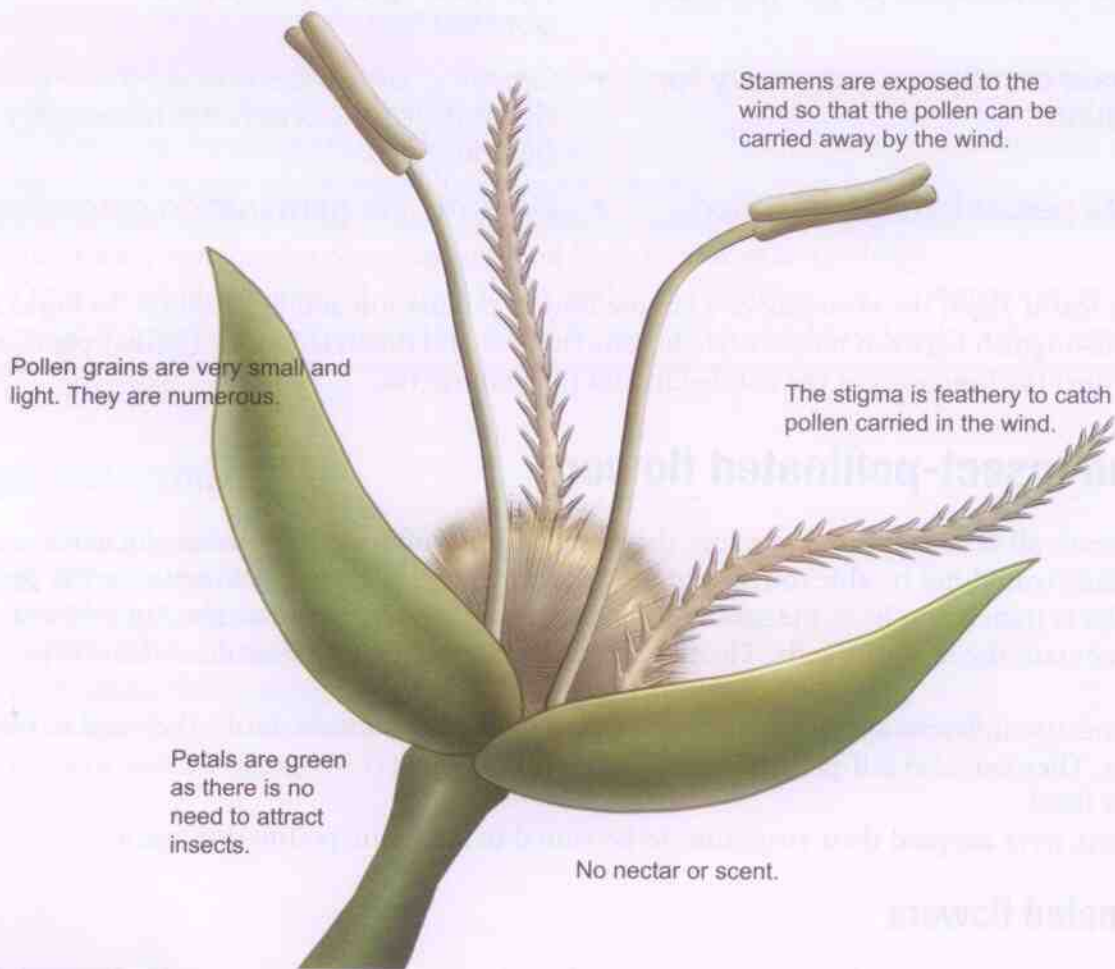


Figure 7.1 Wind-pollinated flower structure

Insect-pollinated flowers

Insect-pollinated flowers are specialised to attract insects. Flowers have adaptations such as large colourful petals, scent or sweet **nectar** to attract insects and birds.

Anthers have sticky pollen grains that stick to the insect's body. The stigma is also sticky so that when the insect lands or brushes past it, the pollen grains on the insect's body stick to the stigma. The stigma and anthers are located inside the flower.

Word help

nectar: a sugary fluid made by flowers to attract insects so pollination can occur

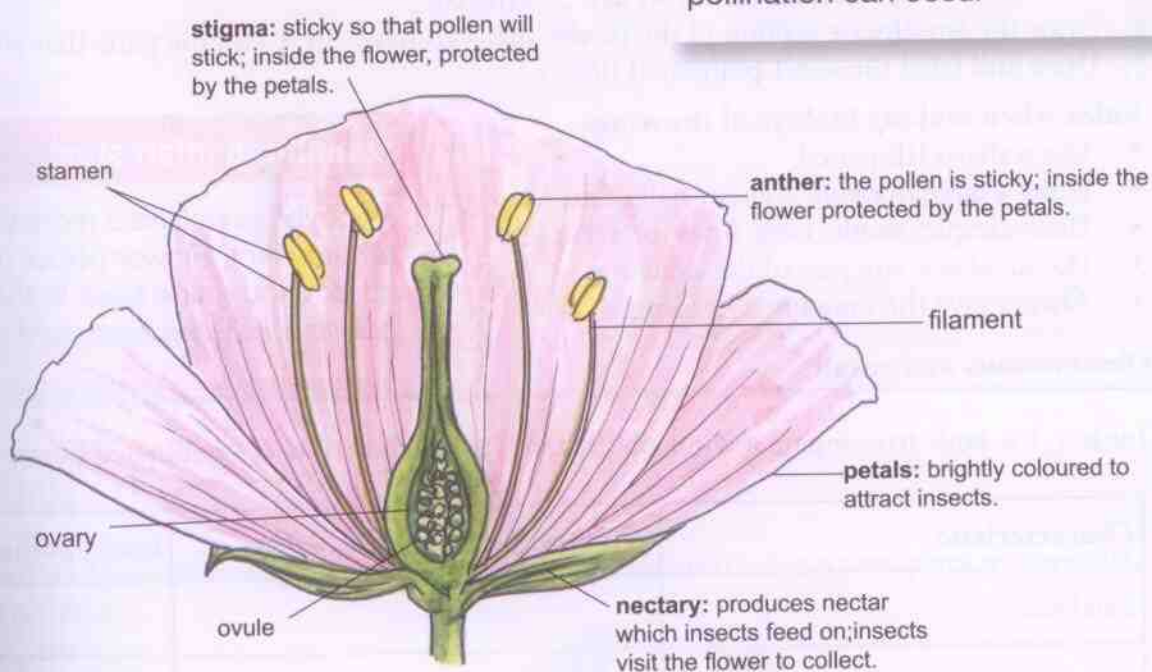


Figure 7.2 Insect-pollinated flower structure

Activity 7.1 Practical task

Examining the structure of wind- and insect-pollinated flowers Work with a partner

Aim: To investigate the structure of wind and insect-pollinated flowers

You will need:

hand lens
sharp cutting tool
variety of insect- and wind-pollinated flowers

Safety

Take care with sharp cutting tool.

Method

Follow the instructions.

1. Collect a variety of insect- and wind-pollinated flowers.
2. Group the flowers into two groups, wind pollinated and insect pollinated.
3. Use the hand lens to identify as many parts of both flowers as you can. Discuss the functions of the parts with your partner.
4. Compare the flowers looking at: petal size; colour; shape and length of anthers and filaments; **shape, size and stickiness of stigma**; and smell. Describe only the characteristics that you observe on your flowers. Copy and complete Table 7.1.
5. Cut the insect-pollinated flower in half lengthwise.
6. Draw the lengthwise section of the flower that you have cut. Label the parts that you can see.
7. Draw and label the wind-pollinated flower.

Rules when making biological drawings

- Use a sharp HB pencil.
- Make a large drawing – about half of an A4 page.
- Draw simple, single, clear lines for accuracy.
- Do not shade any part of the drawing.
- Always give the drawing a heading.

Something interesting

Honeybees pollinate more than any other insect. Flower colour markings attract and guide bees to them for pollination. Bees like bright colours.

Observations and results

Table 7.1 A table to compare a wind-pollinated flower with an insect-pollinated flower.

Characteristic	Wind pollinated	Insect pollinated
Petal size		
Colour		
Smell		
Stigma		
Anther and filaments		

Questions

Answer the questions.

Indicate whether each of the following statements is true or false? Only write TRUE or FALSE.

1. The stamens of insect-pollinated flowers are outside the flower.
2. Some insect-pollinated flowers have nectar.
3. Wind-pollinated flowers are usually very small.
4. Insect-pollinated flowers have coloured petals.
5. A wind-pollinated flower's stigma is inside the flower.

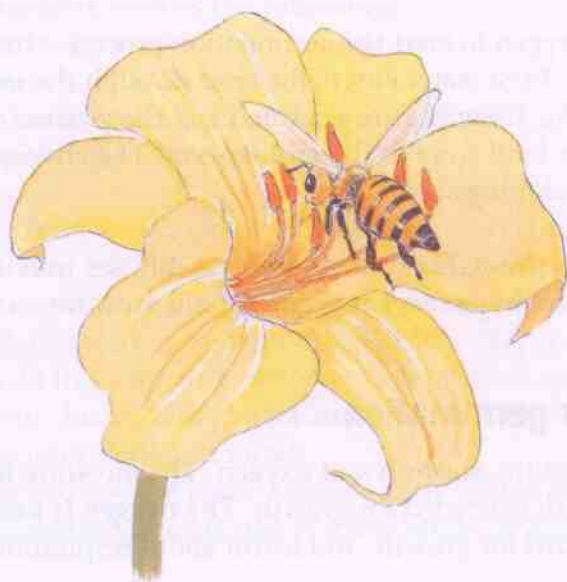


Figure 7.5 A bee pollinating a flower

Germination

After pollination and fertilisation seeds form in the ovules of the ovary. For a new plant to grow the seed needs to germinate.

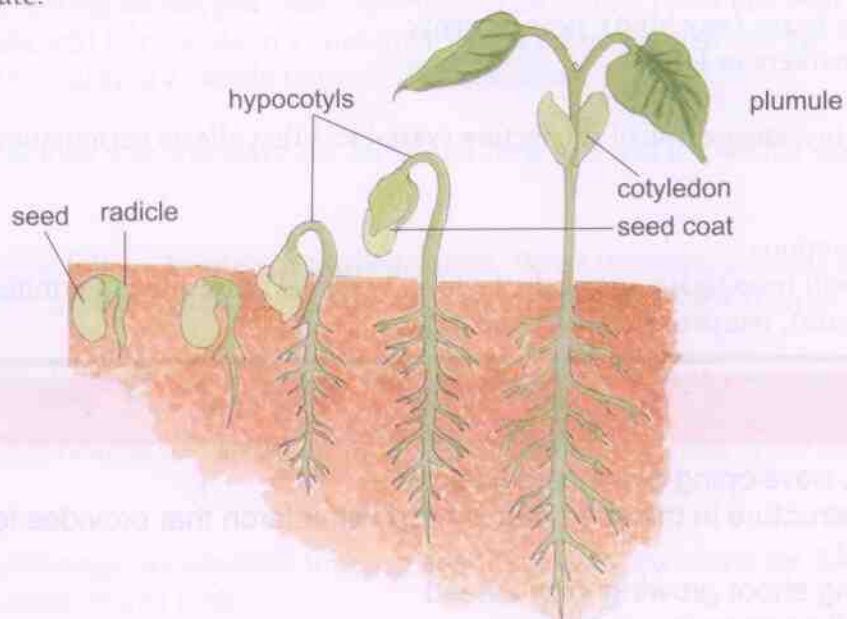


Figure 7.3 Germination of a bean seed

The process of germination

Seeds need moisture, warmth and oxygen to start the germination process. Once they have these requirements germination can begin. First water enters the seed through the **micropyle**. The seed swells and the seed coat (testa) breaks. Enzymes are activated and the process of respiration begins. The enzymes start to break down the food store in the **endosperm**. The endosperm is filled with starch that provides food for the developing seed.

Once the food source (starch, protein or oils) becomes soluble it diffuses into the seed **embryo**. The embryo starts to grow. The **radicle** and **hypocotyl** appear first and grow into a young seedling. Look at Figure 7.3.

The conditions necessary for germination

For a seed to germinate it needs moisture, warmth and oxygen. The moisture is needed to soften the seed coat and growth. The warmth is needed for growth. The oxygen is needed for respiration. Respiration provides the energy needed for growth. You learnt about respiration in Form 2.

Activity 7.2 Practical task

The conditions necessary for germination Work in a group

Aim: To investigate the effect of temperature, moisture, light and oxygen on germination

You will need:

- suitable containers (petri dishes, clear plastic cups, cut-off bottoms of plastic water bottles or small glass jars)
- cotton wool or paper towel
- seeds such as beans (any kind), peas or maize
- permanent markers or labels

Each group will investigate one of the factors (variables) that affects germination.

Method

Follow the instructions.

1. Each group will investigate one of the factors (variables) that affects germination. Choose from water (moisture), temperature, light and oxygen.

Word help

embryo: young, developing plant inside a seed

endosperm: a structure in the seed that is filled with starch that provides food for the developing seed

hypocotyl: young shoot growing from a seed

micropyle: small opening in a seed coat.

radicle: young root growing from a seed

Ask yourselves questions such as the following:

- Will seeds grow in dry conditions?
 - Will seeds grow in a refrigerator?
 - Do seeds grow faster under light or dark conditions?
 - Will seeds grow in a sealed jar?
2. The method depends on which factor you decide to investigate. Discuss how and what your group is going to do.
 3. Keep a day-to-day record, in order to monitor progress of the growth of the seeds. During the investigation check the seeds every day to make sure they have the right conditions.
 4. You are required to do the following: write an investigative question, formulate a hypothesis, design a test for your theory, record data and draw conclusions.
 5. Keep the following information in mind:
 - The more seeds you use the more accurate your results will be. At least 10 seeds should be used for each condition. If only one or a few seeds are used, some may be diseased or not normal.
 - When testing for a condition, all other conditions should be kept constant.
 - Use the same number of seeds in each container.
 - For each experiment there should be a **control experiment**.
 - If you are testing whether moisture is needed, the control experiment would be one without moisture.
 - If you are testing for temperature one group of seeds should be in warm conditions. The control experiment should be in the cold.
 - If you are testing for light, one group of seeds should be exposed to light and for the control, the other group of seeds placed in a cupboard, where there is no light.
 - If you are testing for oxygen your experiment can be in a room but your control experiment will have seeds in a container with a lid or a sealed bell jar. Oxygen can be removed by burning a candle in the sealed container.
 6. Consider for how long you will leave the seeds and how you will know that the seeds have germinated?
 7. Design a method to follow during your investigation. Write the steps in point form. You must have a set of clear, precise instructions. The steps should follow a logical order.
 8. Design a table for the results. Include a column for the percentage germination.
 9. Perform the experiment as set out in you method. Record your results at the end of the experiment.
 10. Calculate the percentage germination for each condition that you tested for. Enter the percentage into your results table.

11. How to calculate the percentage germination.

It doesn't matter how many seeds you have.

$$\text{percentage} = \frac{\text{number of seeds germinated}}{\text{total number of seeds}} \times 100$$

12. Write a conclusion for your experiment.

Questions

Answer the questions.

1. Did your seeds germinate? If not, suggest why?
2. Discuss which parts of the investigation went well.
3. Did you have any problems during this investigation? Why was this?
4. What would you change?
5. What could you have done better?

Indigenous knowledge

Nzara yapera is a traditional maize-seed variety thriving in a changing climate, such as in Chibhememe, in the lowveld region of Zimbabwe. The name means "hunger is gone" and is traditionally used by the Shangaan people. *Nzara yapera* is a short-season maize variety that grows better in dry conditions.

Summary

- Characteristics of insect-pollinated flowers:
 - large, coloured/scented petals
 - some produce nectar
 - stamens inside the flower
 - carpel inside the flower
 - the stigma is sticky
- Characteristics of wind-pollinated flowers:
 - small or no petals
 - stamens outside the flower on long filaments
 - carpel with feather stigma
 - numerous, small, light pollen
- Germination is the start of the growth of the young embryo inside the seed into a plant.
- The conditions necessary for germination are:
 - moisture (water)
 - warmth (suitable temperature)
 - oxygen
- The percentage of germination is calculated by:
$$\frac{\text{number of seeds germinated}}{\text{total number of seeds used}} \times 100$$

Topic assessment

Answer the questions.

1. a) Label the parts: A to C.

[3]



Figure 7.4 Lengthwise section through a flower

- b) What is the function of B? [1]
 c) What structures are found in B that form seeds? [1]
 d) Which part of this flower attracts insects? Explain why? [2]
 d) Is this plant an example of an insect- or wind-pollinated flower? Give a reason for your answer. [2]
2. Describe how flowers have adaptations to promote pollination by each of the following pollinating agents: [3]
 a) wind [3]
 b) insects [3]
3. a) State three factors that are necessary for germination. [3]
 b) Which factor starts the process of germination? [1]
4. A learner put 25 seeds in a container. She placed the seed between cotton wool soaked in water and put them in a dark, warm cupboard. She labelled them A. She put another 25 seeds in the same cupboard and covered the seeds with water. She labelled this container B. These are her results:

	A	B
Number of seeds	25	25
Number of seeds germinated	21	7
% germination		

- a) Calculate the percentage (%) germination for A and B. [2]
 b) (i) Suggest why the percentage germination in Group B was so low? [1]
 (ii) Which factor might have reduced the chance of germination? [1]
 c) Why is it better to use 25 seeds instead of only five? [1]
 d) What evidence in a seed shows that germination has started? [1]

Total marks [25]



1. The first step in the process of...
2. The second step is to...
3. The third step is to...
4. The fourth step is to...
5. The fifth step is to...
6. The sixth step is to...
7. The seventh step is to...
8. The eighth step is to...
9. The ninth step is to...
10. The tenth step is to...



11. The eleventh step is to...
12. The twelfth step is to...
13. The thirteenth step is to...
14. The fourteenth step is to...
15. The fifteenth step is to...
16. The sixteenth step is to...
17. The seventeenth step is to...
18. The eighteenth step is to...
19. The nineteenth step is to...
20. The twentieth step is to...

Table with 3 columns and 4 rows		
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36
37	38	39
40	41	42
43	44	45
46	47	48
49	50	51
52	53	54
55	56	57
58	59	60
61	62	63
64	65	66
67	68	69
70	71	72
73	74	75
76	77	78
79	80	81
82	83	84
85	86	87
88	89	90
91	92	93
94	95	96
97	98	99
100	101	102

Topic 8 Reproductive systems in humans

Learning objectives and activities

Learning objectives	Activities
• State the functions of the male and female reproductive systems	• Drawing the male and female reproductive systems
• Describe the structure and functions of sex cells	
• Draw and label the sex cells	• Drawing the structure of sex cells
• Describe the route of the sperm from the testis to the ovum after ovulation	
• Define fertilisation	
• Describe the menstrual cycle	• Illustrating the menstrual cycle
• Describe the role of the placenta	• Watching video simulations of the placenta
• Identify the substances exchanged in the placenta	• Listing the substances exchanged in the placenta

In Form 1 you learnt about the signs of puberty. In Form 2 you learnt about the parts and the functions of the male and female reproductive systems.

The function of the reproductive system is to produce offspring. Humans reproduce by **sexual reproduction**.

The male reproductive system

The male reproductive system is adapted to produce sperm, to deliver the sperm to the female and to produce hormones that regulate the male reproductive system.

The male reproductive organs include the penis and two testes, a sperm duct (vas deferens), scrotum and urethra. The testes produce millions of sperm cells each day. The testes are in the scrotum, which hangs outside the body. The scrotum helps keep the testes at a constant temperature for the development of the sperm.

Word help

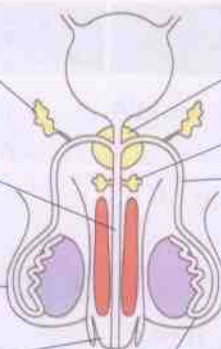
sexual reproduction: reproduction through the fusing of gamete: male sex cell (sperm) and female sex cell (ova)

Seminal vesicle – a pair of glands that secrete a thick sugary fluid containing fructose, which the sperm uses as a food source.

Urethra – a duct leading from the bladder through the penis and opens to the outside at its tip. This opening is called the urogenital opening. The urethra transports urine and semen out of the body at different times.

Scrotum – a double-layered pouch of skin that covers and protects the testes. The temperature in the scrotum is lower than in the body, which is necessary for producing sperm.

Penis – the penis consists of erectile tissue that fills with blood to make the penis hard. The tip of the penis is slightly enlarged and is called the glans penis. The glans is covered by a loose skin called the foreskin. Sometimes it is surgically removed in an operation called a circumcision.



Epididymis – this is a coiled tube that lies at the back of the testis, where sperm are temporarily stored until they are mature.

Prostate gland – a gland that secretes an alkaline fluid that will neutralise any acidic urine or vaginal secretion during sexual intercourse. It also helps sperm to move faster.

Cowper's gland – a pair of glands that secrete mucus to lubricate the opening at the tip of the penis during ejaculation.

Sperm duct (vas deferens) – a duct that carries sperm from the epididymis to the penis.

Testis (pl.: testes) – there are two oval-shaped testes, which hang outside the body in the scrotum. Each testis contains seminiferous tubules that produce the sperm. These tubules open into a larger coiled tube, the epididymis, lying outside at the back of each testis. As sperm are produced, they move from the testis to the epididymis to be stored. Between the seminiferous tubules are masses of interstitial cells that produce the hormone testosterone. Testosterone stimulates the production of sperm, maintains the reproductive organs and causes the secondary sexual characteristics of men.

Figure 8.1 Frontal view of the male reproductive system.

The female reproductive system

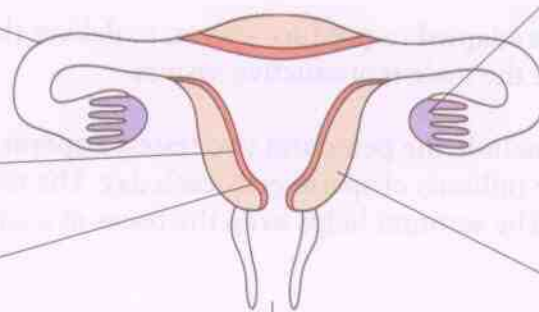
The female reproductive system is adapted to produce egg cells (ova), to receive sperm and to hold and nourish the growing baby during pregnancy until birth. The female reproductive organs include two ovaries (sing. ovary), two Fallopian tubes (oviducts), a uterus and a vagina. The ovaries produce the ova (sing. ovum).

The female reproductive system also produces hormones that control the menstrual cycle.

Fallopian tube or oviduct – this is a narrow tube with muscular walls that are lined with hair-like structures called cilia. It is about 10 cm long. The ova from the ovaries are moved along the Fallopian tube by the cilia and by contraction of the muscular walls.

Endometrium – the layer of tissue that lines the inner wall of the uterus.

Cervix – the lower part of the uterus that connects the uterus with the vagina. It contains glands that secrete mucus into the vagina.



Ovary – there are two ovaries that lie in the abdominal cavity of a female. Ovaries produce ova. The hormone oestrogen is secreted by the ovaries. Oestrogen regulates the menstrual cycle, maintains a pregnancy and causes the secondary sexual characteristics of women.

Uterus – also known as the womb, is a thick-walled muscular organ.

Vagina – the vagina is a muscular canal that runs from the cervix and opens to the outside of the body. At the opening of the vagina are folds of skin called the labia. Between the labia is a small, roundish structure called the clitoris.

Figure 8.2 Frontal view of the female reproductive system

Activity 8.1 Work with a partner

Answer the questions

1. Describe two ways in which the male reproductive system is adapted to its function.
2. Describe two ways in which the female reproductive system is adapted to its function.
3. Draw and label a diagram of the male reproductive system.
4. Draw and label a diagram of the female reproductive system.

The structure and function of sex cells

Human males produce sperm cells (spermatozoa) and females produce egg cells (ova).

A sperm cell has a head, a middle section and a tail. The head contains a nucleus with the heritable characteristics (DNA) of the father. There are many mitochondria in the middle section of a sperm cell. These provide energy so the sperm can swim towards the ovum. Sperm cells are very small compared to the ova of a female.

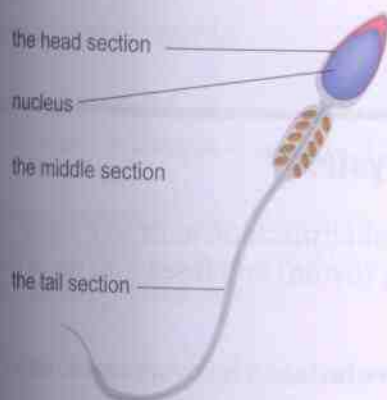


Figure 8.3 The structure of a sperm

An ovum has a nucleus and cytoplasm, and is surrounded by a cell membrane. The nucleus contains the heritable characteristics (DNA) of the mother. The cytoplasm stores food for the embryo to develop and grow.

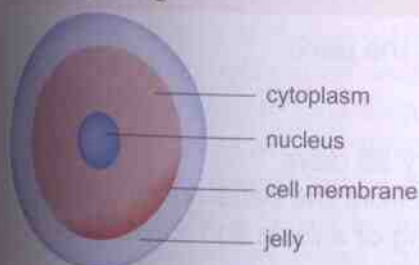


Figure 8.5 The structure of an ovum



Figure 8.4 A micrograph of a sperm

Word help

heritable: characteristics capable of being passed from parents to offspring



Figure 8.6 A micrograph of an ovum

Activity 8.2 Practical

Differences between sperm and ova Work with a partner

Aim: To investigate the similarities and differences between a sperm cell and an ovum

You will need:

- bioviewer
- sperm cell slide
- ovum slide

Method

Follow the instructions

1. Look at the bioviewer slide of a sperm cell or the micrograph of a sperm cell in Figure 8.4.
2. Look at the bioviewer slide of an ovum or the micrograph of an ovum in Figure 8.6.

Questions

Answer the questions.

1. State one similarity between a sperm and an ovum.
2. State two visible differences between a sperm and an ovum.

Draw and label.

- a) a sperm
- b) an ovum.

The movement of sperm through the reproductive system

For reproduction to be successful a sperm must reach an ovum and fertilisation must take place.

Fertilisation takes place when a single sperm enters a mature egg (ovum) and **fuses** with the ovum to form a **zygote**. Fertilisation takes place in the oviduct.

An ovum is released from an ovary every 28 days. This is called **ovulation**. The ovum moves into the oviduct.

For the sperm to reach the oviduct it travels from the testis, where it is produced, to the epididymus where it is temporarily stored. During **ejaculation** thousands of sperm are released in a liquid called **semen**. The sperm moves through the sperm duct and out through another tube that runs through the penis, called the urethra.

Word help

ejaculation: when the sperm leaves the male body through the penis

fuse: join; combine

fertilization: the fusing of sex cells to form a zygote

ovulation: the release of an ovum from the ovary once every 28 days

semen: thick, whitish liquid that contains sperm and spurts from the penis during ejaculation

zygote: a fertilised ovum. **zygote:** a cell formed by the joining of a male and a female sex cells

The sperm is deposited in the vagina of a female during sexual intercourse. From the vagina the sperm swims through the cervix, which is the opening of the uterus. From the uterus the sperm swims up into the oviduct to fuse with the ovum. Fertilisation takes place and a zygote forms. The zygote immediately starts dividing into new cells. At the same time it moves towards the uterus. Once this ball of dividing cells reaches the uterus it attaches to the spongy tissue of the endometrium that lines the uterus wall. This process is known as **implantation**. The female is now pregnant and should not menstruate again until the baby is born.



Figure 8.7 Electron micrograph of a sperm cell and an egg cell during fertilisation

The menstrual cycle

A female is born with hundreds of immature ova (egg cells) within her ovaries. When she reaches puberty, these ova begin to mature. One ovum is released from an ovary every 28 days.

The menstrual cycle takes place in females and each cycle lasts about 28 days. The function of the menstrual cycle is to cause ovulation and to prepare the uterus for implantation. The menstrual cycle is controlled by hormones.

There are two main hormones that control the menstrual cycle: oestrogen and progesterone.

Oestrogen causes the uterus to form a special nourishing lining consisting of blood and blood vessels. This lining is the endometrium and prepares the uterus for implantation. The oestrogen level rises during the first part of the cycle and then falls. This rise and fall contributes to ovulation.

Progesterone is released during the second half of the cycle. It keeps the lining inside the uterus in place. If implantation occurs the lining will be kept in place until the birth of the baby. If implantation does not occur, the levels of progesterone drop. This causes the uterus lining to break down and pass out of the body through the vagina. This is **menstruation**. A female begins to menstruate during puberty. Blood and small

Word help

implantation: when the ball of dividing cells attaches to the uterus lining after fertilisation

menstruation: the loss of the uterus lining once every 28 days after puberty. It passes out through the vagina

blood vessels from the special uterus lining come out of her vagina and we say her period has started. Look at Figure 8.8 to see all the events that occur during one menstrual cycle.

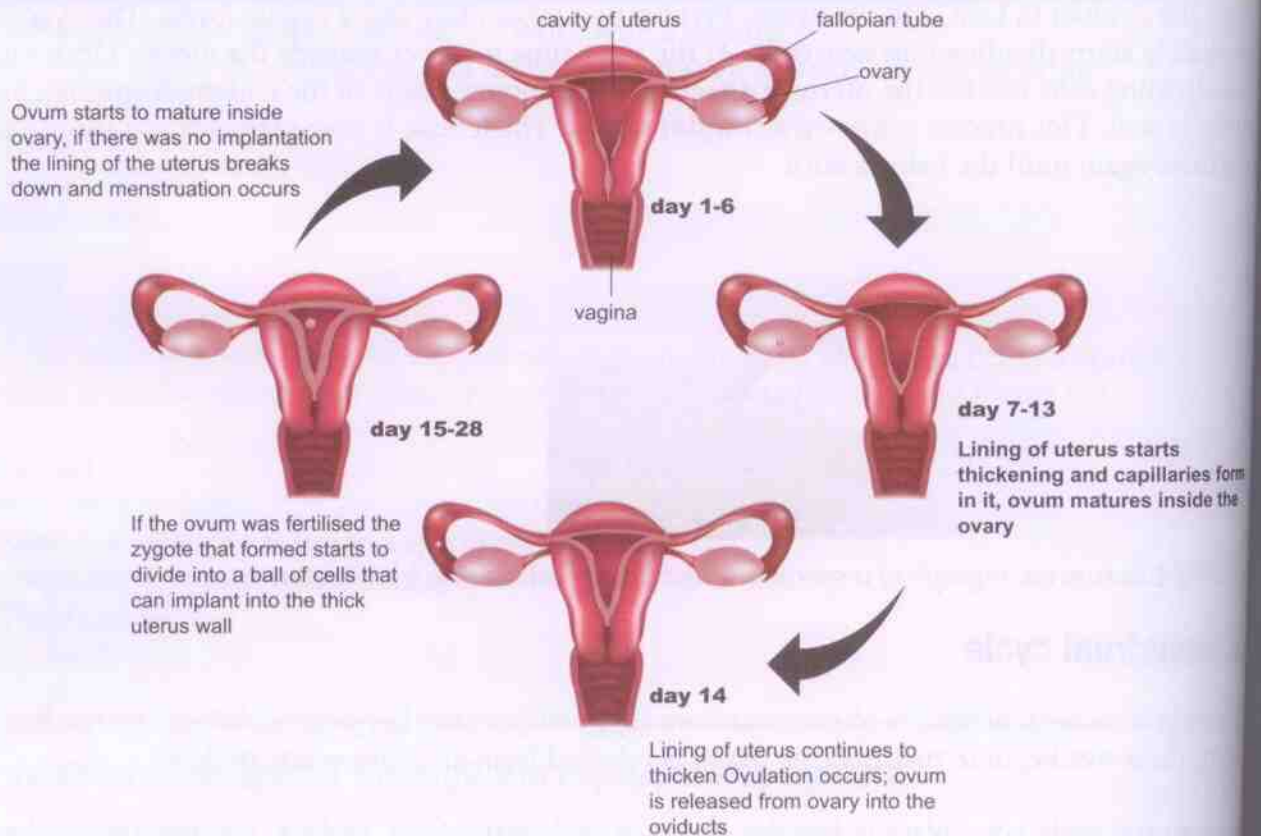


Figure 8.8 The events taking place during one menstrual cycle

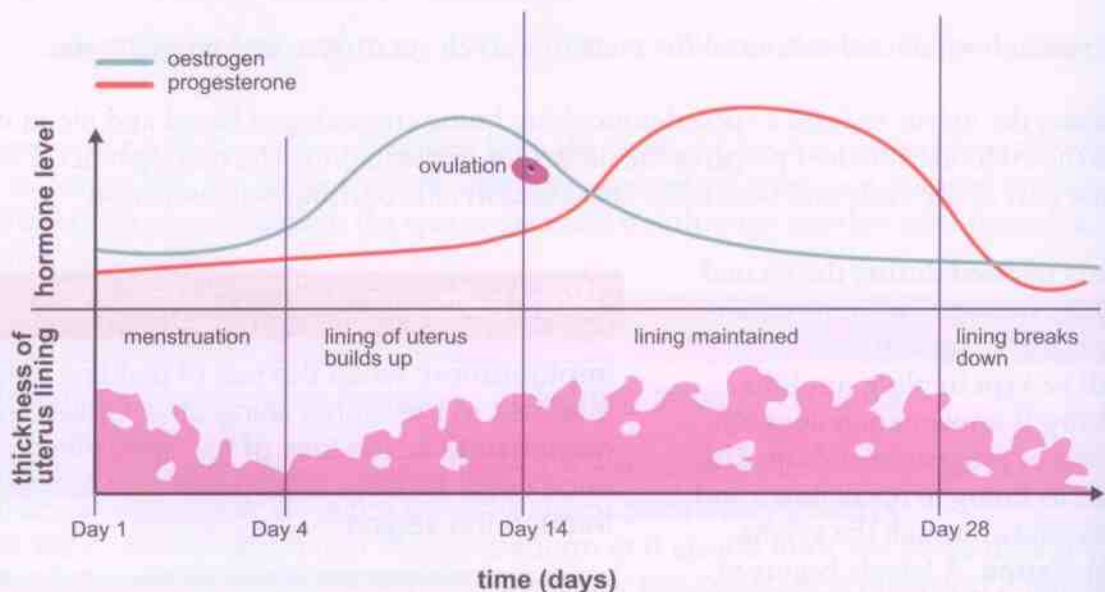


Figure 8.9 A graph showing the hormone levels and the thickness of the uterus lining during one menstrual cycle

Activity 8.3

Answer the questions

1. a) Describe what happens between days 1 and 5. b) What causes these events to happen?
b) On which day does ovulation usually happen?
2. Use the graph in Figure 8.9 to answer the following questions.
 - a) Name the hormone that causes the uterus lining to develop.
 - b) On which day does this hormone in a) reach a peak?
 - c) When does the progesterone hormone level rise?
 - d) What happens to the uterus lining if implantation does not take place?
 - e) What causes the event in d) to happen?

The role of the placenta

When an ovum is fertilised the **zygote** starts to divide to form a ball of cells while moving down the oviduct towards the uterus. Later, this ball of cells will become a baby. The uterus lining is ready to receive the ball of cells. The ball of cells attaches to the uterus lining during implantation. The ball develops into an **embryo**.

Once the zygote has implanted itself into the endometrium of the uterus, a special disc-like structure forms to feed it. This structure is called the **placenta**.

The placenta is a filter between the mother's blood and the baby's blood. In the placenta substances such as nutrients, oxygen and waste products move between the blood of the developing embryo and the mother. However, their blood never mixes. The developing embryo is attached to the placenta by the **umbilical cord**.

The placenta stops many harmful substances that could be in the mother's blood from reaching the foetus. But it cannot protect the baby against nicotine from cigarettes, alcohol, other drugs and certain microorganisms such as syphilis. You will learn more about syphilis in Topic 9 Health and diseases.

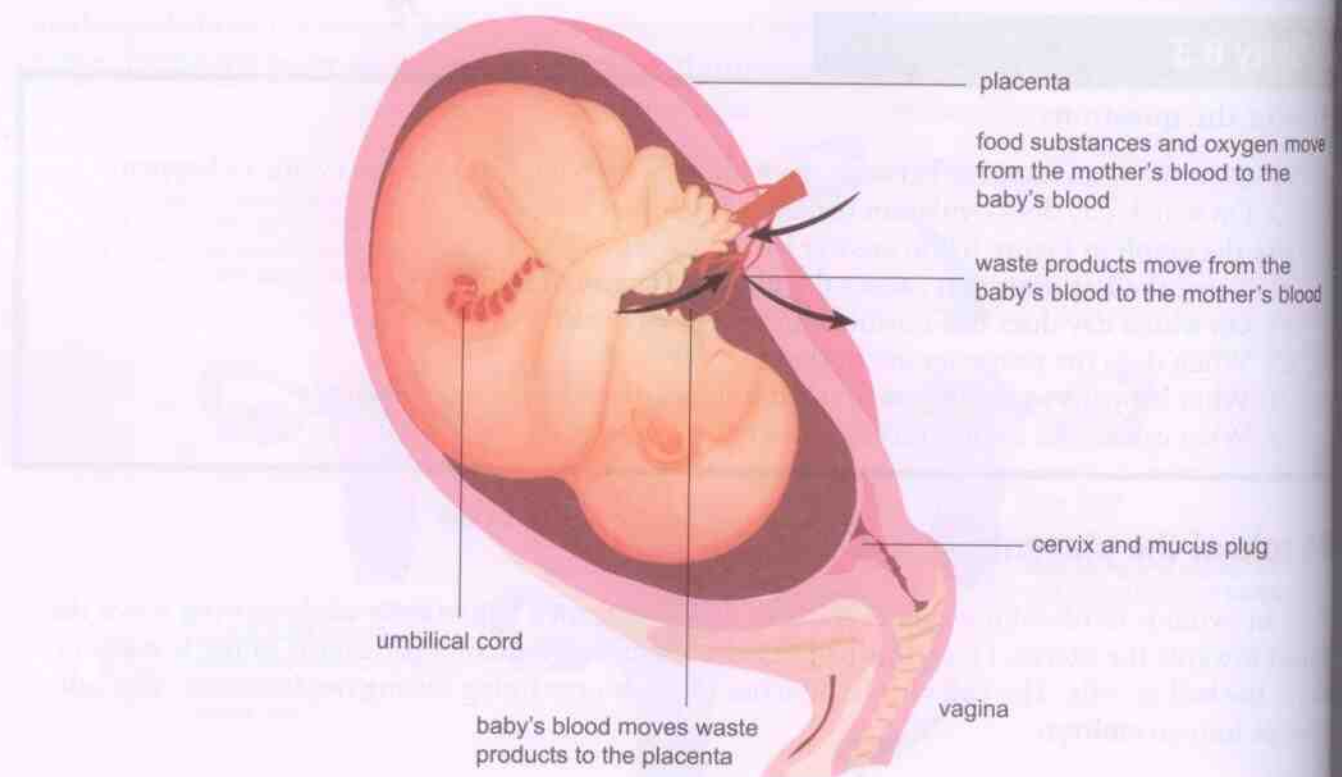


Figure 8.10 Foetus developing in the uterus

As the embryo grows into a baby, the placenta also grows. After week eight the embryo is called a foetus. Figure 8.10 shows the foetus and placenta.

The placenta develops many blood vessels and membranes to create a large surface area for the exchange of substances between mother and baby.

Nutrients like glucose, amino acids, fatty acids and glycerol, vitamins and minerals can enter from the mother to the foetus. Antibodies and oxygen also enter the foetus through the placenta. The placenta also removes waste products like carbon dioxide from the foetus.

Something interesting

About 20 percent of a pregnant woman's blood supply goes through the placenta every minute. When a baby is born, the placenta will be about 20 cm in diameter and weigh about 1.3 g.

Activity 8.4 Work with a partner

Answer the questions.

1. Where does the placenta attach?
2. What is the name of the cord that arises from the placenta?
3. Can the placenta remove waste products like alcohol from the mother's blood?
4. Can the placenta provide a foetus with protection from certain diseases? How?
5. Why should a mother not smoke during pregnancy?

Summary

- The male and female reproduction systems have various functions.
- Male sex cells or sperm are very small and have a tail for movement.
- Female sex cells or ova are much larger and have a food store. Ova cannot move by themselves and are moved along by cilia in the oviduct.
- The route that sperm travel to get to the female ovum is as follows: testes, epididymis, vas deferens, urethra, vagina, cervix, uterus, oviduct.
- Fertilisation is the moment when a sperm and ovum fuse (join) to form a zygote.
- The menstrual cycle is on average 28 days and ovulation is on day 14.
- The hormone oestrogen helps to build the uterus lining.
- Progesterone maintains or keeps the lining in place if an ovum is fertilised.
- Ovulation is the release of an ovum from the ovary.
- Menstruation is the loss of blood and fluid from the old uterus lining, which passes out through the vagina on average every 28 days.
- The placenta attaches the developing embryo/foetus to the mother's uterus.
- The baby receives soluble nutrients, oxygen and antibodies from the mother's blood and passes carbon dioxide and other waste out to the mother's blood supply.

Topic assessment

Answer the questions.

1. Figure 8.11 shows the human female reproductive system.

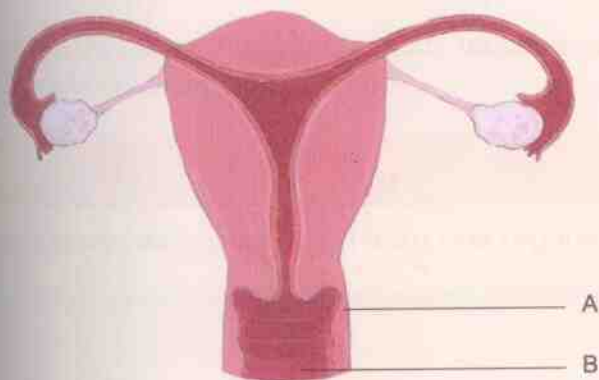


Figure 8.11 The female reproductive system

- a) Name the parts labelled A and B. [2]
 - b) Name the structure in which ova are made? [2]
 - c) In which part does fertilisation take place? [1]
 - d) Name the structure where a baby develops. [1]
2. a) List two structural differences between a sperm cell and an ovum. [2]
 - b) State one functional difference. [1]
 - c) Name the product of fertilisation. [1]
3. a) Name the structures that sperm must pass through to reach the female. [4]
 - b) Describe the structure and functions of the placenta. [6]

Total marks [20]

Topic 9 Health and diseases

Learning objectives and activities

Learning objectives	Activities
List sexually transmitted infections (STIs)	Discussing STIs
Describe the signs, symptoms, effects and causes of: gonorrhoea, syphilis, chancroid, genital herpes	Identifying the signs, symptoms, effects of STIs. Watching videos of STIs
State the control methods and treatment	List the causes of STIs
Describe the signs, symptoms and causes of malaria, typhoid, Ebola and cholera	- Using charts on the signs and symptoms - Discussing the causes
Explain how the diseases are treated	Outlining the disease-control methods
Describe the effects of tobacco smoking on health	Discussing the effects of smoking
Describe the effects of excessive alcohol consumption	Identifying the effects of excessive consumption of alcohol
Explain the effects of the use of Mandrax and Cannabis	Listing the effects of Mandrax and Cannabis
Outline the effects of breathing solvents	Explaining the effects of breathing solvents

In Form 1 you learnt the importance of being healthy and maintaining personal hygiene and you described methods in which diseases are **transmitted**. In Form 2 you learnt about the causes of diseases and all about bilharzia.

Sexually transmitted infections (STIs)

Sexually transmitted infections (STIs) is a group of diseases transmitted or spread through sexual intercourse or other forms of sexual contact. You learnt about sexual intercourse in Topic 8 Reproductive systems in humans.

There are many diseases that are sexually transmitted. You will focus on four namely: gonorrhoea, syphilis, chancroid and genital herpes.

Word help

transmitted: being passed on from one organism or human to the next organism or human

Activity 9.1 Work with a partner

Answer the questions

Examine Table 9.1 which shows the causative agent, signs, symptoms, effects and treatment of four STIs.

Table 9.1: Summary of four common STIs

Gonorrhoea	
Causative agent	Bacteria – <i>Neisseria gonorrhoeae</i>
Signs Symptoms	• Unusual discharge. Can be greenish yellow or a whitish discharge from the penis or vagina • Painful or swollen testicles (male) or lower abdomen (females) • A burning, painful feeling when urinating
Effects	Can prevent people from having children It can cause death if left untreated
Treatment	Antibiotics
Chancroid	
Causative agent	Bacteria – <i>Haemophilus ducreyi</i>
Signs Symptoms	• Ulcers on male and female reproductive organs • Pain when urinating and passing anal waste
Effects	Scars left from the sores
Treatment	Antibiotics, surgery to drain ulcers.
Genital herpes	
Causative agent	Virus called herpes simplex. The virus can be inactive for long periods of time but can still be passed on, which makes it very dangerous
Signs Symptoms	• Sometimes one or two blisters that break open to produce painful sores occur on reproductive organs • Sometimes no symptoms or fever, feeling sick, body aches like flu with the appearance of blisters • Symptoms appear at various time intervals
Effects	Painful blisters on reproductive organs that can spread to other parts of your body
Treatment	There is no cure but symptoms can be controlled by anti-viral drugs

Syphilis – three stages over a long time

Causative agent	Bacteria
Signs Symptoms	• Can imitate the signs and symptoms of many diseases • 1st stage: Sore or lesion mostly on or in the reproductive organs • 2nd stage: Rash on body – hands and feet; more sores, weight loss, fever, sore throat • 3rd stage: circulatory system problems, mental problems
Effects	If left untreated, severe body system damage and death
Treatment	Antibiotics in early and middle stages

Answer the questions.

There are many similarities among the STIs.

1. Which parts of the body are affected in all STIs?
2. Which two STIs can cause death?
3. a) Which disease do you think is the most dangerous?
b) Give a reason for your answer.
4. Write a short paragraph on why you think it is important for you to learn about STIs?

Prevention of STIs

The most effective way of preventing an STI is **abstinence**. Other ways of limiting your exposure, is to be in a **monogamous** relationship with one person that you trust and know. Using condoms during sexual intercourse limits the transfer of STIs. It, however, does not give a person complete protection.

Fortunately many STIs can be treated with **antibiotics**. This is especially true if they are discovered early. If you think that you may have an STI, you should see a doctor as soon as possible. If there is even a slim chance that you might have contracted an STI, DO NOT have intercourse with anyone. Prevention is still the best course of action to take.

With some STIs, doctors test and treat everyone that has had sexual contact with the patient. This is called contact tracing. This is done to limit the spread of the disease.

Word help

abstinence: not having sexual intercourse

discharge: released from

monogamous: being in a long-term relationship with only one person

antibiotics: type of antimicrobial drug used in the treatment and prevention of bacterial infections



Figure 9.6 Ways in which a person cannot get a STI

Dangerous infectious diseases in Africa

You will focus on four dangerous diseases in Africa – including Zimbabwe. You will learn about malaria, typhoid, Ebola and cholera. **Risk factors** for typhoid, Ebola and cholera include poor sanitary conditions and poor hygiene.

Malaria, typhoid, Ebola and cholera signs and symptoms

Table 9.2: Some dangerous infectious diseases

Disease	Signs	Symptoms
Malaria	Repeated shaking with a fever; severe sweating; anaemia; diarrhoea	Headache, nausea, muscular pain
Typhoid	Fever, diarrhoea.	Pain in abdomen; poor appetite; headache; lack of energy
Ebola	Red eyes; rash; diarrhoea; severe loss of weight, bleeding	Chest pain and coughing; stomach ache
Cholera	Rapid diarrhoea; vomiting	Nausea and dehydration

Table 9.3: Causes of certain infectious diseases

Disease	Cause	Treatment	Control
Malaria	Plasmodium parasite – carried by the Anopheles mosquito. The mosquitoes breed in still standing water.	A combination of medicines also referred to as ACT	Using insecticide-treated nets (ITNs); insect repellants; gauze on open windows; mosquito nets; wearing long-sleeved shirts and long trousers; taking anti-malaria medication; clearing stagnant water and long grass in urban areas
Typhoid	Bacterium – spread in contaminated water or food	Antibiotics and rehydration therapy .	Wash before eating or preparing food and after using the toilet; avoid drinking untreated or contaminated water; sewage needs to be disposed of in a responsible way; a vaccine is available
Ebola	Ebola virus. Spreads by physical contact with contaminated body fluids of an infected person	No cure; the symptoms are treated by rehydration therapy and blood transfusion; early treatment increases a person's chances of survival	Avoid contact with bodily fluids of an infected person; practise strict body hygiene; when there is an outbreak of Ebola people with the disease are isolated to prevent the spread of the disease
Cholera	Bacterium in contaminated food or water	Rehydration therapy; antibiotics	Using a safe piped water supply; boiling unsafe water for at least a minute before use; correct use of toilets; hygienic food preparation; basic body, food and water hygiene

Word help

rehydration therapy: treatment to replace fluids in a person that is dehydrated due to e.g. diarrhoea

risk factors: factors that will increase the chances of a person getting a disease

Something interesting

In Zimbabwe, cases and deaths from malaria have decreased from about two million cases and 5 000 deaths per year in 2000, to below 400 000 cases and less than 300 deaths per year. This is a result of the government and the people using methods to prevent the spread of this disease.



Figure 9.1 A female *Anopheles* mosquito

Activity 9.2 Work with a partner

Answer the questions.

Look at tables 9.2 and 9.3.

1.
 - a) What is the cause of malaria?
 - b) How is malaria spread from person to person?
 - c) List three ways in which people can reduce the chance of being infected with malaria.
2.
 - a) Are typhoid and cholera caused by a bacterium or a virus?
 - b) Describe two ways in which the spread of both of these diseases can be reduced.
 - c) What treatments are there for people suffering from typhoid and cholera?
3. Choose one of the diseases above and design an information pamphlet for your community.

Substance abuse

Substance abuse is also known as drug abuse. It is when a person consumes or smokes a harmful substance regularly.

Tobacco smoking

The smoking of any tobacco products like cigarettes or a pipe is very harmful to the human body and can severely affect your health. The smoking of tobacco damages many different parts of the body but especially the lungs. Tobacco products contain toxic substances such as nicotine, carbon monoxide and tar. These substances cause health problems such as emphysema, lung cancer, bronchitis and heart disease. When pregnant women smoke, there is less oxygen available for the development of the foetus. This can cause pregnant mothers who smoke to have babies with a low birth weight.

In addition to the many health problems the chemicals can cause, nicotine is a very addictive substance. This makes it very difficult for people who smoke to give up. The best way to not get addicted is not to start smoking.

Emphysema

Emphysema is a serious disease that makes a person very short of breath. Cigarette smoke irritates the alveoli walls, causing them to burst, which results in huge spaces. You learnt about alveoli in Topic 4. In Figure 9.2 you can see the damaged alveoli. There is a reduction in the surface area in the alveoli for gaseous exchange to occur. People with emphysema are always short of breath and feel as if they are suffocating. Eventually they will not be able to walk and will have to breathe in air from an oxygen tank.

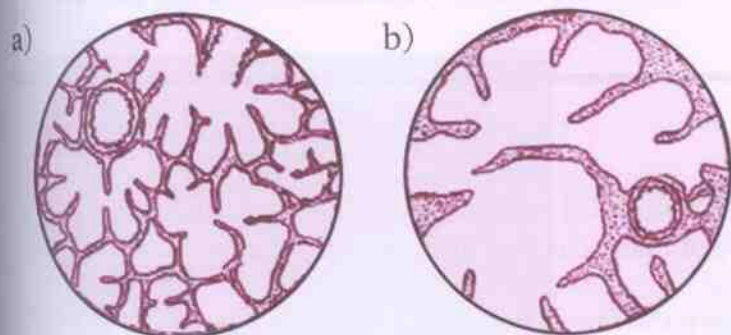


Figure 9.2 Cross section of a) normal alveoli and b) the alveoli of a person with emphysema

Bronchitis

Bronchitis is caused when the bronchi in the lungs become inflamed. The inflammation causes excess mucus to form in the lungs. This leads to painful coughing bouts. People who smoke are more prone to getting bronchitis.

Lung cancer

Just smoking a few cigarettes increases the chances of getting lung cancer. Every time you smoke, the **carcinogens** in the smoke damages the lung tissue. Your body is able to repair this damage but sometimes the cells in your lungs start growing uncontrollably. This causes lung cancer. Tar, a substance found in cigarettes, forms a sticky layer in the lungs. See in Figure 9.3. This sticky layer also damages the lungs so that less oxygen is absorbed.



Figure 9.3 a) the lungs of a non-smoker and b) the lungs of a smoker

Heart disease

Some of the toxic substances in cigarette smoke cause the arteries to narrow, which forces the heart to work harder. This effort increases the blood pressure. This will eventually lead to a weak heart.

Activity 9.3 Work in a group

1. Create a colorful poster with pictures and drawings, warning school children about the dangers of smoking.
2. Do some research on the Internet or the library to find relevant information. You can also use the information from this text book.

Excessive alcohol consumption

Alcohol is a very dangerous substance. It causes many road deaths each year. It can lead to aggressive behaviour, loss of inhibitions and it is the cause of many families breaking up.

Alcohol affects our nervous system. It slows down the rate at which our brain sends and receives nervous impulses. Drinking excessive amounts of alcohol can lead to liver cirrhosis, reduced reaction time and have social implications.

Cirrhosis of the liver

Alcohol is toxic to liver cells, and many heavy drinkers develop cirrhosis, when the liver is so damaged that it cannot function. A person suffering from liver cirrhosis may be tired all the time, bleed easily, have a yellow colour to the whites of their eyes, have a loss of appetite, abdominal swelling and nausea. Eventually toxins will build up in the body and a person could die.

Reduced reaction time

Alcohol affects our nervous system. Because of this we react slower than we normally would. This becomes very dangerous if we drink and drive. Many road accidents are caused because drivers do not stop fast enough or swerve fast enough in dangerous situations. Drivers often ignore the rules of the road when they are driving under the influence of alcohol. Many pedestrians are killed yearly because they step in front of cars when they are drunk.

Social implications

There are many alcoholics in Zimbabwe. An alcoholic is a person who is addicted to alcohol. Being an alcoholic can have far-reaching social implications. It is very hard to keep a job and earn money to support oneself or one's family when a person is an alcoholic.

Alcoholic mothers find it very difficult to look after their children properly. Mothers who drink excessively during pregnancy may have children who suffer from Foetal Alcohol Syndrome (FAS). Children with FAS can have a low intelligence, be very aggressive and not develop properly. These children need constant care and will never become responsible adults.

Mandrax and Cannabis

The use and abuse of drugs is a problem in many countries, including Zimbabwe. People use drugs for many different reasons. Many people become addicted to drugs and so cannot stop using them.

Mandrax and Cannabis are drugs used in Zimbabwe. They go under different names. Mandrax can be called Mandies, MX, Buttons, Whites and other names. Mandrax is a **depressant drug**.

Cannabis can be called weed, grass, dagga, pot and other names but its true name is marijuana. It is a depressant drug but it can also be a **hallucinogenic** drug. It is very common in Zimbabwe.

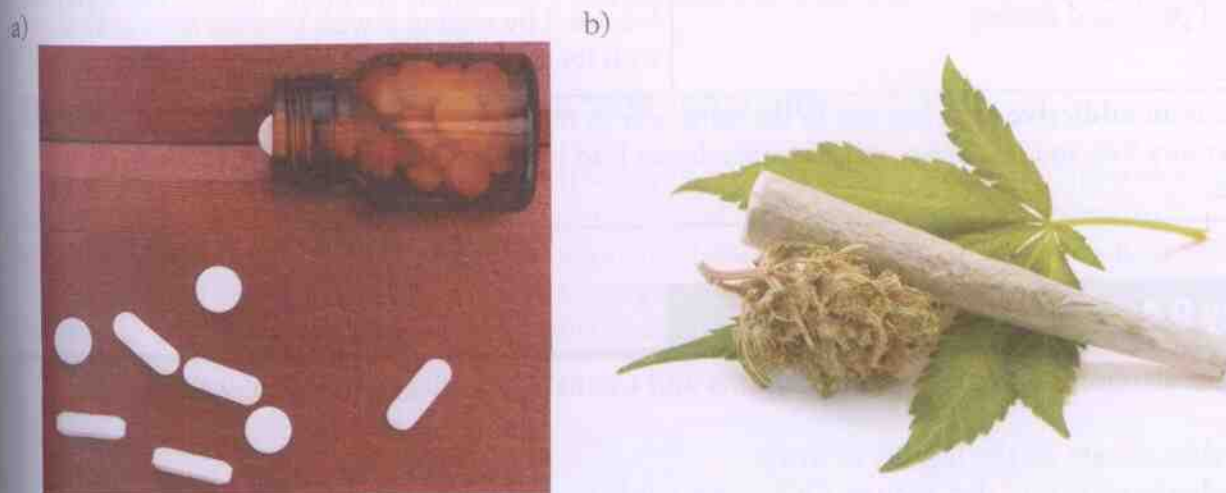


Figure 9.4 a) Mandrax and b) Cannabis

Word help

hallucinogen: drug or chemical that changes a person's thinking and awareness of reality

depressant drug: makes a person feel calm and relaxed – even sleepy

Table 9.4 Summarise the effects, form and use of Mandrax and Cannabis

Mandrax

Effects	Form and use
It makes a person feel relaxed, peaceful, calm and happy. Everything feels good! However, it can also make users feel sick and sleepy. Some people feel confused and aggressive when it wears off. Regular users will lose weight.	It comes as a coloured or white tablet. It is sometimes crushed and mixed with Cannabis and smoked. It can be swallowed or injected.
Regular mandrax users need it more and more as they become addicted. It is very difficult to stop and is an expensive habit. When people use it for a long time, their mental and physical health weaken and they can go into a coma or die.	

Cannabis

Effects	Form and use
A feeling of dreaminess, bliss, happiness, hallucination, strange awareness of surroundings OR feelings of panic and anxiety.	Cannabis is usually green in colour and can be moist, dry or compressed . It often looks similar to tobacco. It is used by mixing it with tobacco for smoking or with food for eating.
Cannabis is an addictive drug but not in the same way as Mandrax. When used for a long time, it can cause memory loss and less awareness of reality. It can lead to emotional, social and physical health problems.	

Activity 9.4

At present alcohol is a legal drug but Mandrax and Cannabis are illegal in Zimbabwe.

Have a class debate on the legality of drugs.

1. Divide the class into two groups. One group could argue that: alcohol, Mandrax and Cannabis should ALL be illegal; and the other group that alcohol, Mandrax and Cannabis should ALL be legal.

Word help

addictive: a strong need to regularly have something

coma: to be unconscious for a long time

solvent: a liquid which dissolves a substance – for example acetone, alcohol

Breathing solvents

Liquids like alcohol, petrol and acetone dissolve substances that are mixed with them. They are called **solvents**. You have learnt about solvents in chemistry. Solvents give off a vapour that we can smell.

Some people sniff or inhale the vapour or fumes from a substance like glue or paint. Often, they spread the solvent on a cloth (sometimes, they put the cloth into a paper bag) and then breathe in the vapours. The solvent is breathed in and it has an effect on the brain for a short time.

Effects

Sniffing solvents causes a brief feeling of happiness. However, it also causes lack of control of the muscles, speech and reason, as well as dizziness and sleepiness.

When used for a long time, it is possible to die due to a lack of breathing or damage to the liver, lungs, heart and brain. Sniffing harmful substances may also cause very bad depression, which may lead to suicide.

Activity 9.5 Case study

Work on your own and then with a partner.

Read the story about Sipho on your own. Then discuss with your partner and answer the questions.

The story of a 12 year old boy who took inhalants

In the story *No turning back* by South African writer Beverley Naidoo, 12-year-old Sipho escapes from his violent step-father and arrives in Johannesburg. There he meets up with other gangs of children who live on the streets – two in particular, Joseph and Jabu – and who struggle in their fight against hunger, cold and danger. On his first night sleeping on the streets, Sipho thinks of his mother. He imagines he will never see her again and begins to cry. One of the other boys, Joseph, hears him and tries to comfort him.

It was Joseph. The thing being pushed into Sipho's hands was the bottle of *iglue* (glue).

'The first night is always bac. Me, I was only eight and I was crying all night because by Ma, she didn't want me any more ...'

Without stopping to think any more, Sipho put the bottle up to his nose and took a couple of sharp, deep breaths.

'Take some more, man!' whispered Joseph.

Sipho sniffed again, until suddenly his head felt quite light, almost dizzy. He didn't like the feeling but lay down, resting his head again on Jabu's shoulder. With Joseph now on the other side of him, he felt warmer. It was good being close to somebody else.

Joseph's mother hadn't wanted her son and his own Ma didn't care about him anymore. But now he was remembering how good it used to feel when as a very small boy he had slept alongside his grandmother.

Source: Darwin Namibia Env. Grade 6 and Beverly Naidoo (extract from No turning back pages 24 and 25)

1. How did Joseph comfort Sipho?
2. Do you think this was the right thing for him to have done?
3. How else could Joseph have helped Sipho?
4. Suggest an ending to this part of the story. What could have happened to Sipho?



Summary

- There are four main sexually transmitted diseases (STIs).
- Gonorrhoea – caused by a bacterium, treated with antibiotics
- Syphilis – caused by a bacterium, treated with antibiotics
- Chancroid – caused by a bacterium, treated with antibiotics
- Genital herpes – caused by a virus, treated with anti-viral drugs
- STIs can be controlled by not having sex (abstinence), staying with one life partner or by only having safe sex.

- Condoms reduce the transmission of the pathogens and infected people need to trace and inform their contacts.
- Malaria - caused by Plasmodium and transmitted by Anopheles mosquitoes
- Typhoid – caused by a bacterium and spread by contaminated food and water
- Ebola – caused by a virus and spread by contact with an infected person's body fluids
- Cholera – caused by a bacterium and spread by contact with contaminated water
- Smoking tobacco products can result in emphysema, bronchitis, lung cancer, heart problems and low birth weight in pregnant mothers.
- Excessive consumption of alcohol affects the body and can lead to liver damage, social problems and reduced reaction times.
- People can become addicted to Mandrax and Cannabis.
- Breathing solvents like glue and vapours from fuels can lead to a loss of coordination, breathing difficulties, damage to body systems and death.

Topic assessment Work on your own

Answer the questions.

- List two sexually transmitted infections (STIs). [2]
 - Which STI is a long-term infection in stages if not treated? [1]
 - State how STIs can be treated if detected early. [1]
 - Suggest a way in which STIs can be prevented or avoided. [1]
- Compare Ebola and typhoid. Use a table to show one similarity and one difference. [2]
 - Which diseases are spread by unsafe food and water? [2]
 - Suggest how the spread of these two diseases can be reduced. [1]
- Malaria kills many children every year. What causes malaria? [1]
 - How is the disease spread? [1]
 - Suggest how people can protect themselves from being bitten by mosquitos in their homes. [1]
 - What can be done in the local environment to reduce the spread of malaria? [2]
- What kind of drugs are Cannabis and Mandrax? [2]
 - What are the dangers of using either of these drugs? [2]
 - Breathing difficulties, death or a coma result from which kind of drug abuse? [1]

Total marks [20]

Word help

bronchitis: inflammation and blocking of the bronchi, which reduces air flow in and out of the lungs

carcinogens: substances that cause cancer

depressant drug: makes a person feel calm and relaxed – even sleepy

inhalant: a gas or vapour from a solvent that can be breathed in

lesion: an area on the skin that appears abnormal

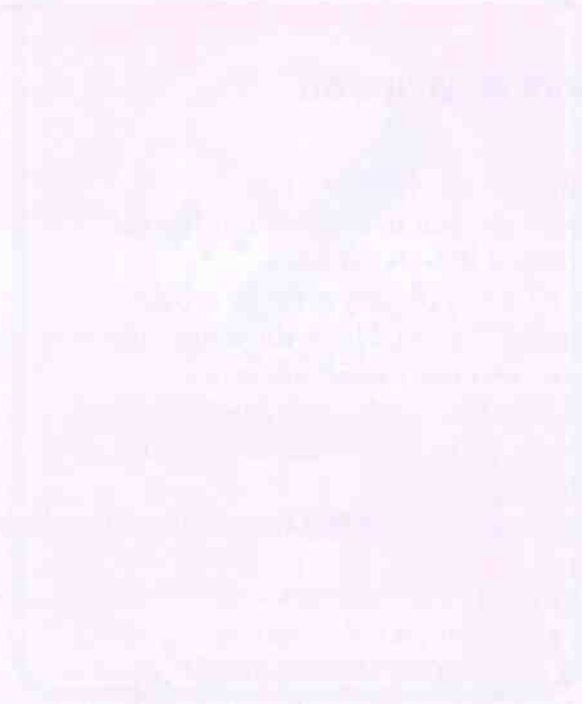
leukaemia: cancer of the blood

monogamous: being in a long-term relationship with only one person

parasite: an organism that lives in or on another and feeds from it

Plasmodium: the name of the parasite that lives in Anopheles mosquitoes and humans

stroke: death of brain cells caused by reduced blood supply



Term

2

Physics



Topic Number	Topic	Learning objectives
Topic 1	Data presentation	Construct a pie chart Interpret and analyse data from pie charts and line graphs
Topic 2	Measurement	Measure physical quantities accurately using appropriate instruments Read instruments' scale to the nearest fraction of the division Determine density of liquids
Topic 3	Force	Weight and mass Define weight, momentum and inertia Distinguish between weight and mass State Newton's laws of motion Calculate force, given mass and acceleration State the application of Newton's laws of motion
Topic 4	Machines	Machines Describe the uses and applications of machines Determine Mechanical Advantage (MA), Velocity Ratio (VR) and efficiency of levers inclined plane, pulleys and gears Explain energy losses in machines Describe ways of improving efficiency in machines